Electronics Circuits I

Chapter 1: Diodes and applications

Dr. Pham Nguyen Thanh Loan

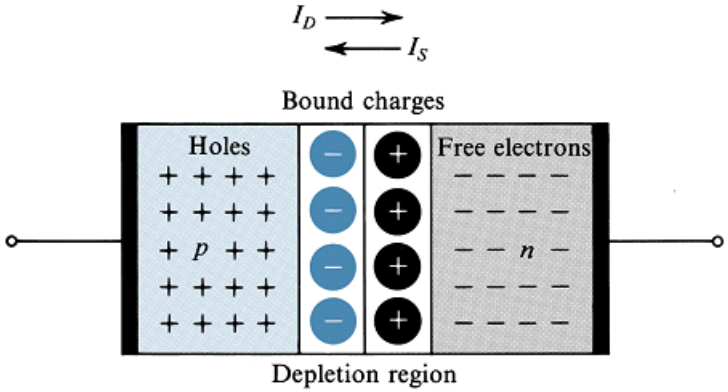
Ha Noi, 21/8/2013

Chapter 1: Diode & applications

Contents

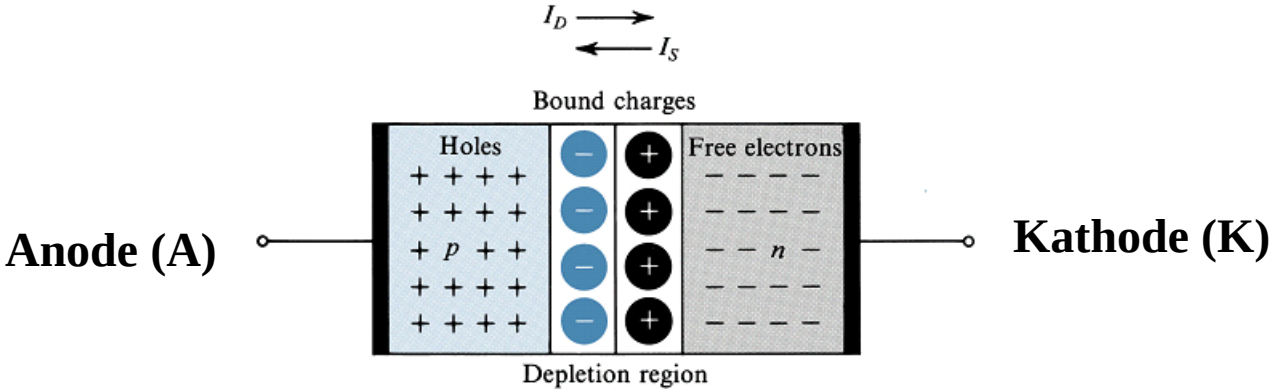
- Diode – Structure, operation and equivalent circuit
- Rectifiers circuits
- Limiters circuits
- Double voltage and so on
- Zener diode and its applications

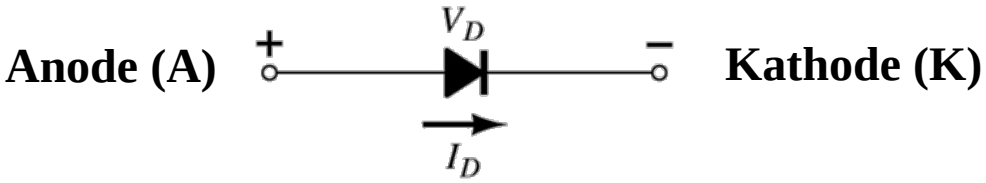
Diode structure



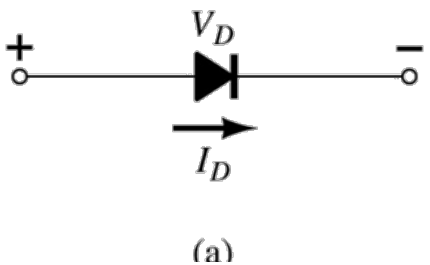
- PN Junction
  - Depletion region: no carriers
  - Ion located at the interfaces with P/N → barrier voltage V_D → drift current I_S (i_{drift})
  - Diffusion phenomenon: flow of charges from high concentration region to lower concentration region → I_D (i_{diff})
  - Equilibrium of PN junction: Is + I_D = 0

PN junction and its symbol: Diode

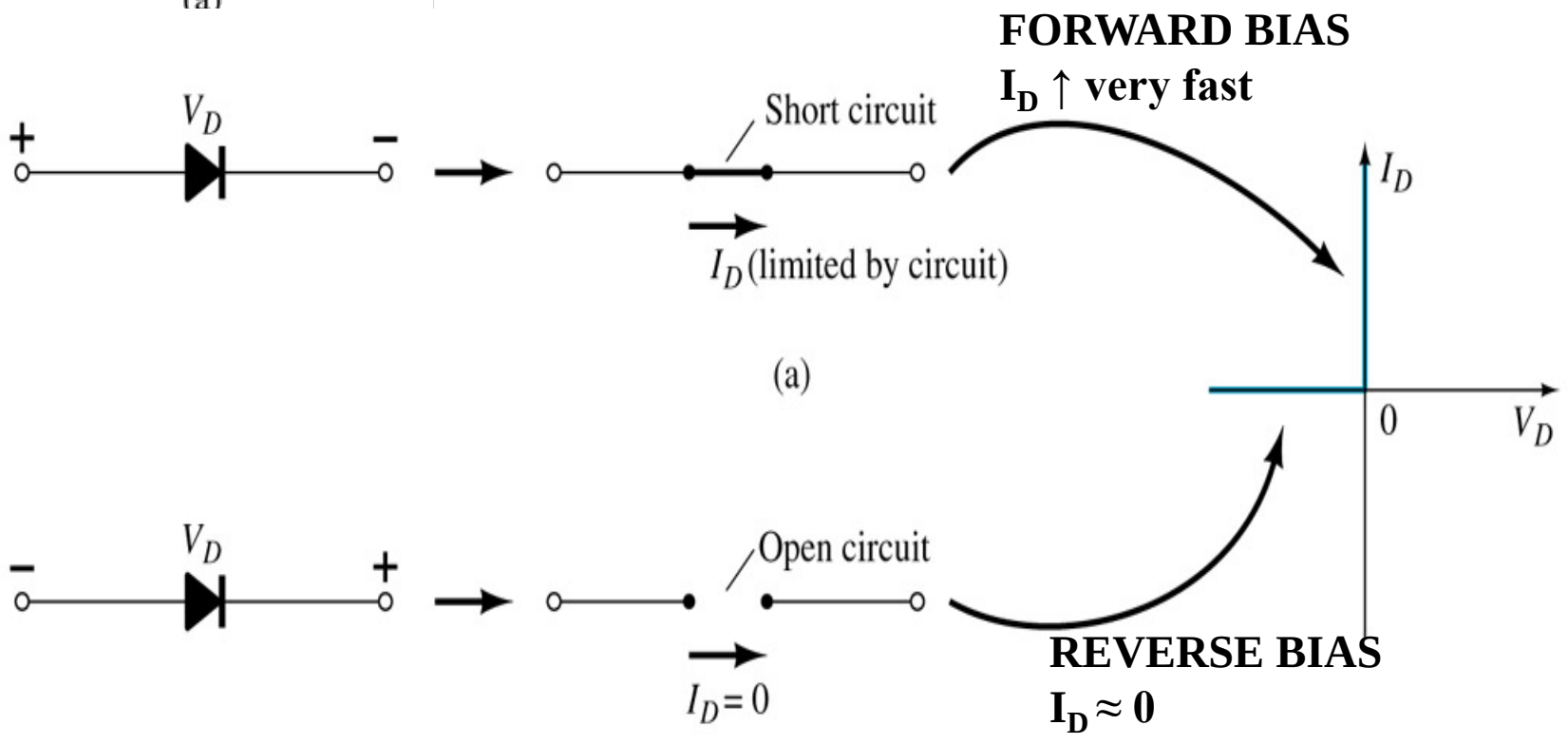




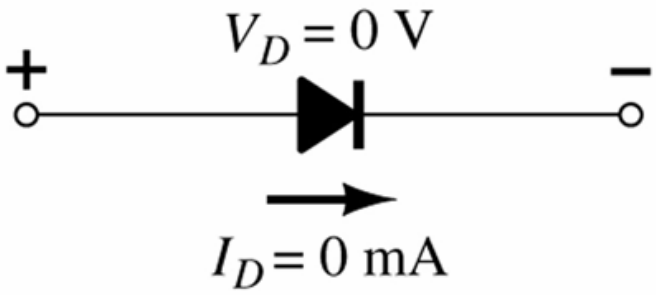
Model of ideal diode

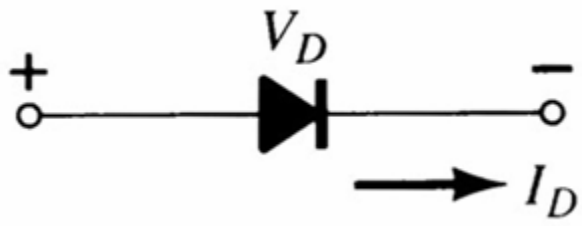


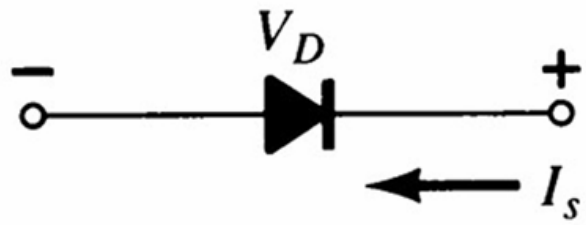
- Two terminals
- Current ON in one direction
- $V_D = 0V$



Model of practical diode (VD ≠ 0)







- FORWARD BIAS: applied voltage > V_D

$V_D \approx 0.7V$ (Si)

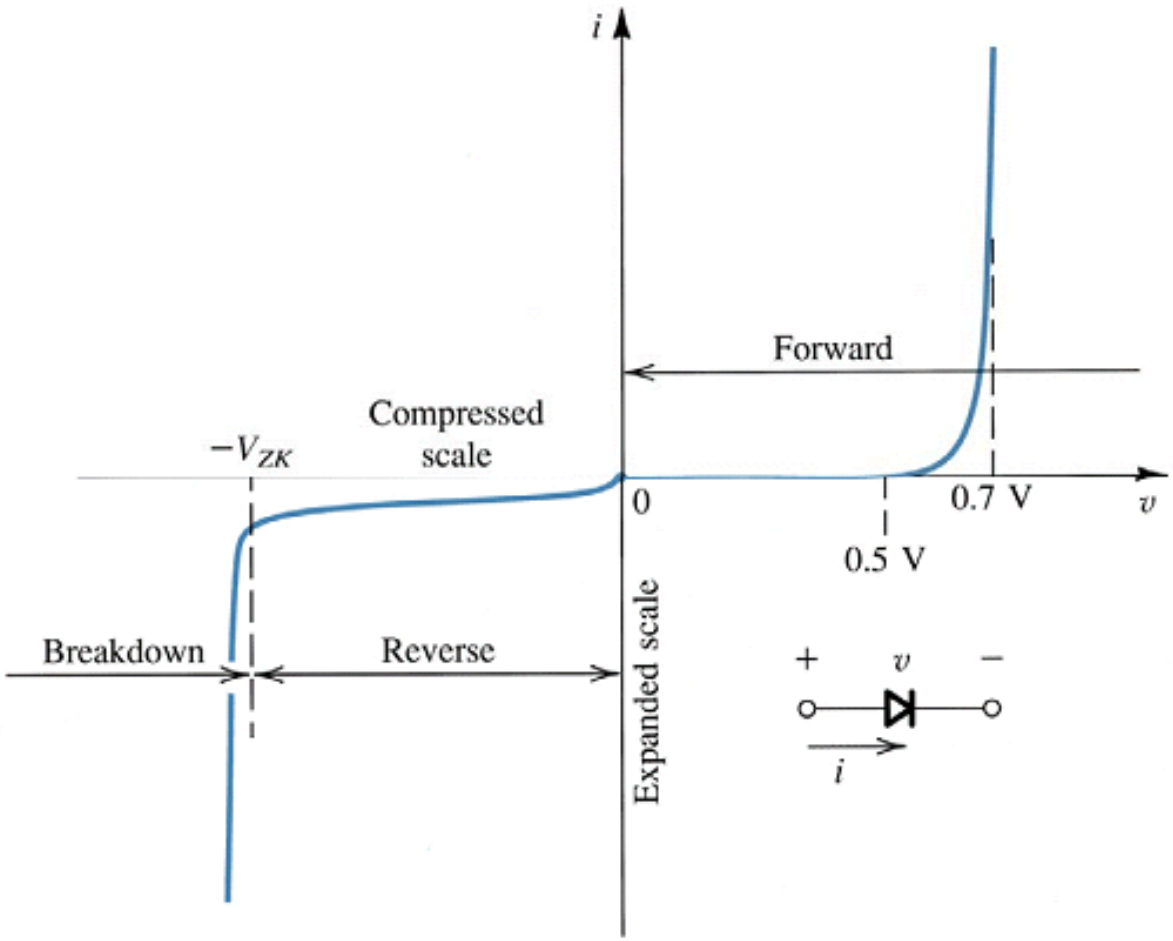
$V_D \approx 0.3V$ (Ge)

➔ Voltage drop on diode = V_D

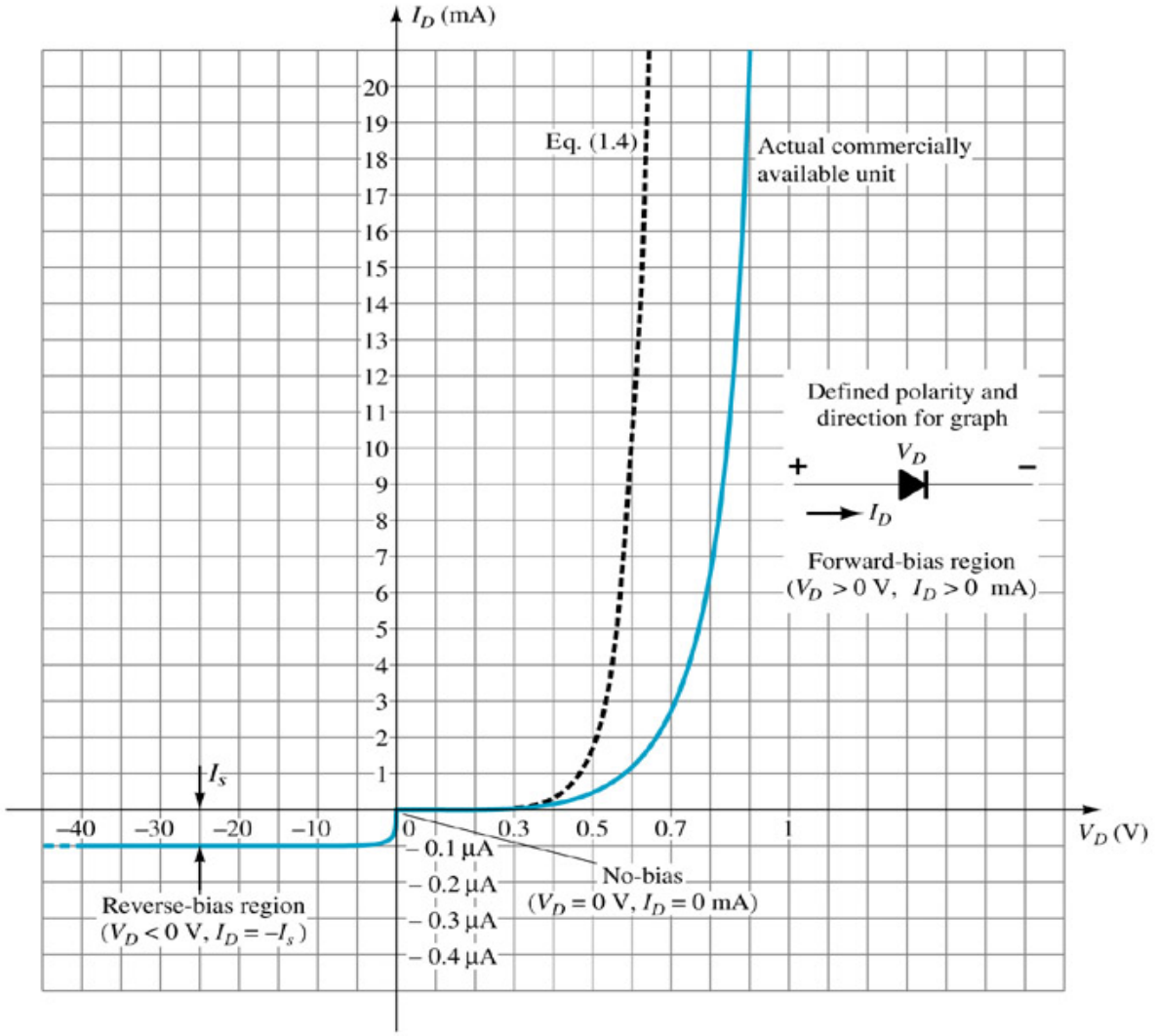
- REVERSE BIAS:
  - Leakage current $I_S \approx 0$

➔ Open circuit

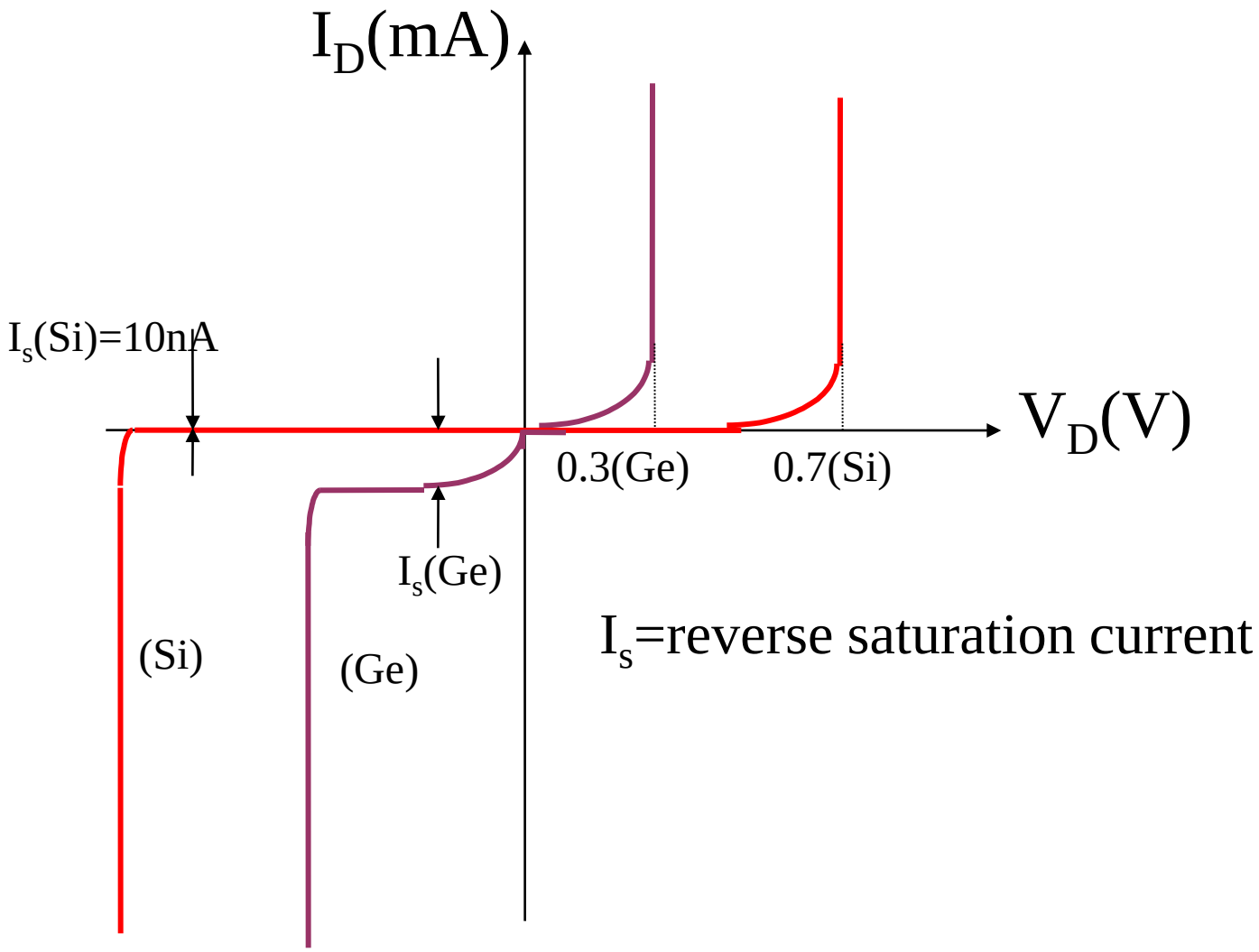
PRACTICAL Diode



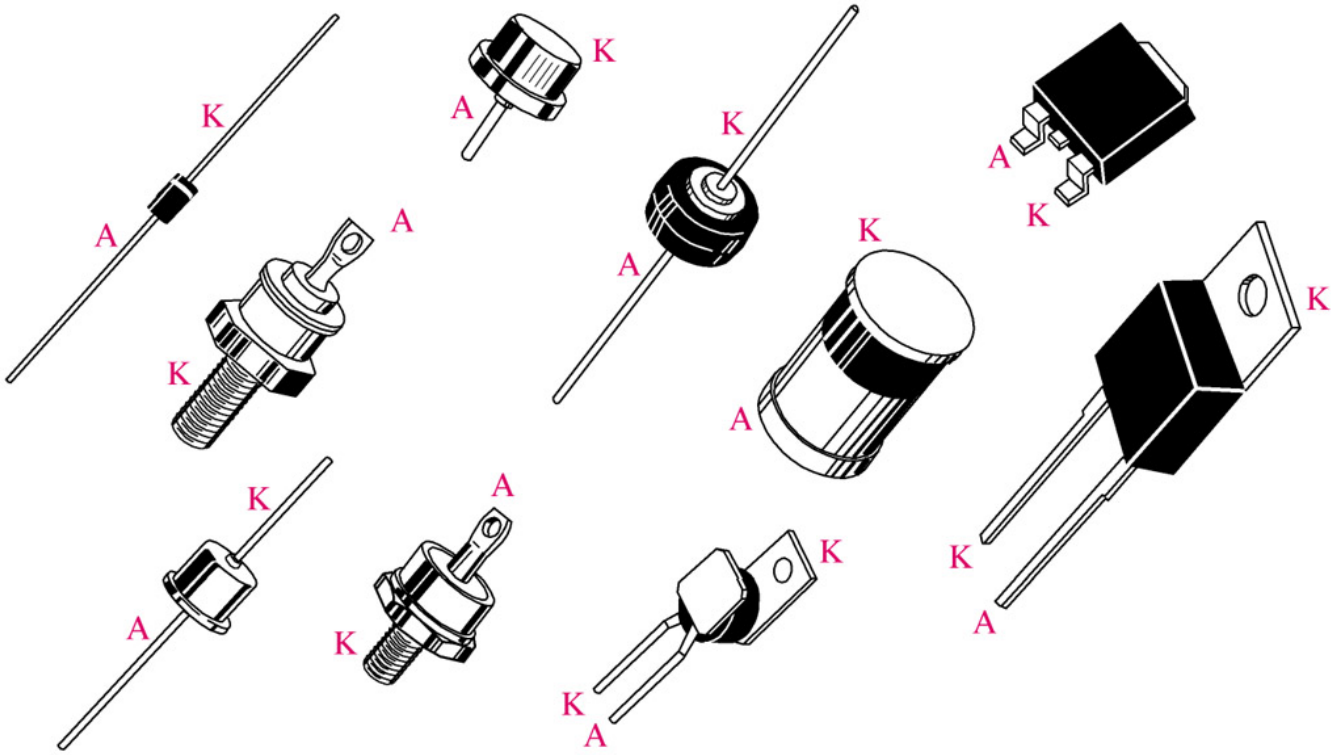
PRACTICAL Diode



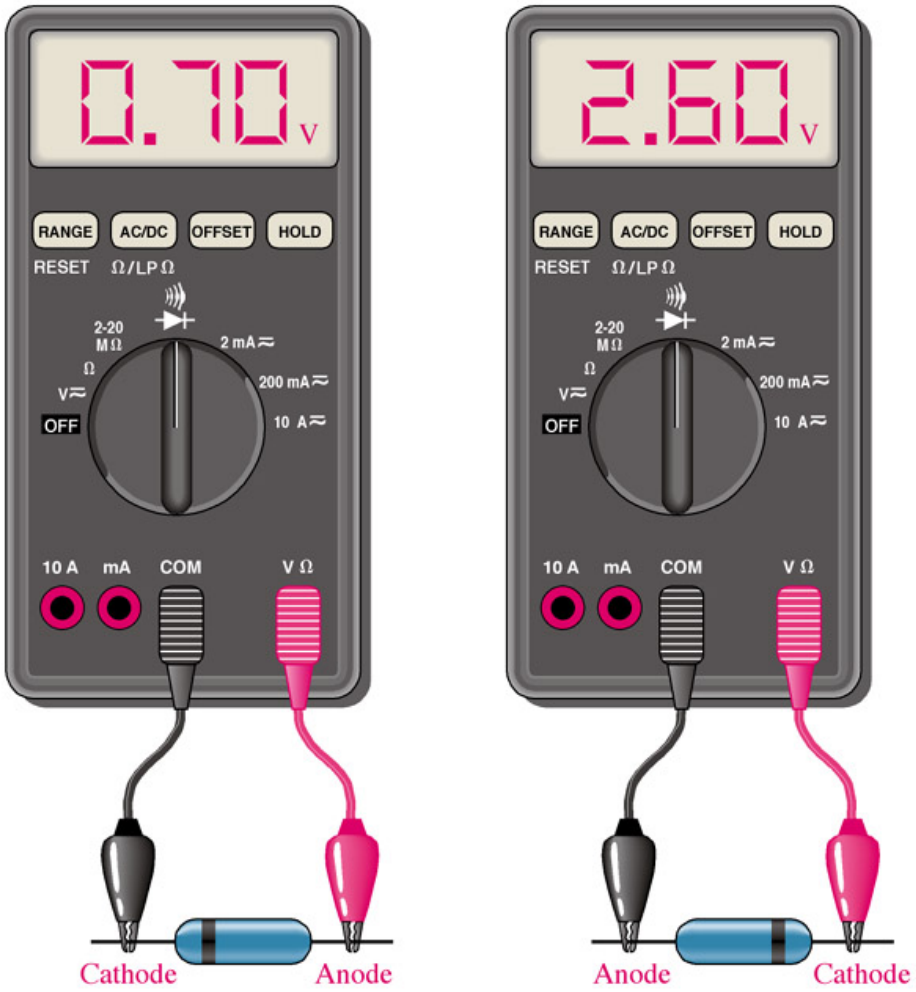
PRACTICAL Diode (VD ≠ 0)



PRACTICAL Diode



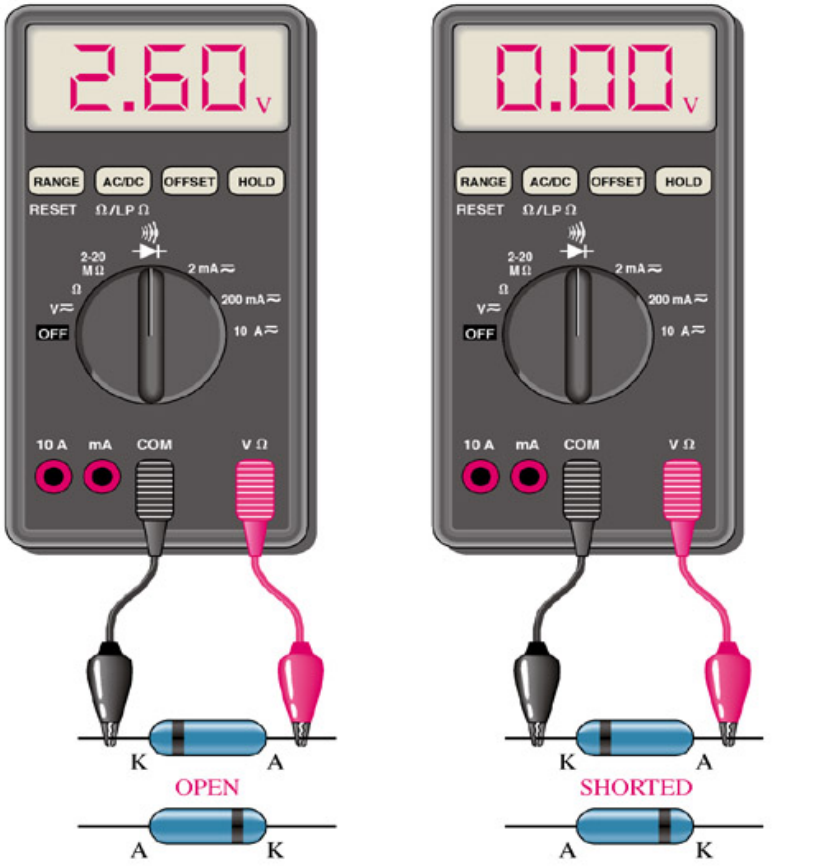
PRACTICAL Diode - multimeter



(a) Forward-bias test

(b) Reverse-bias test

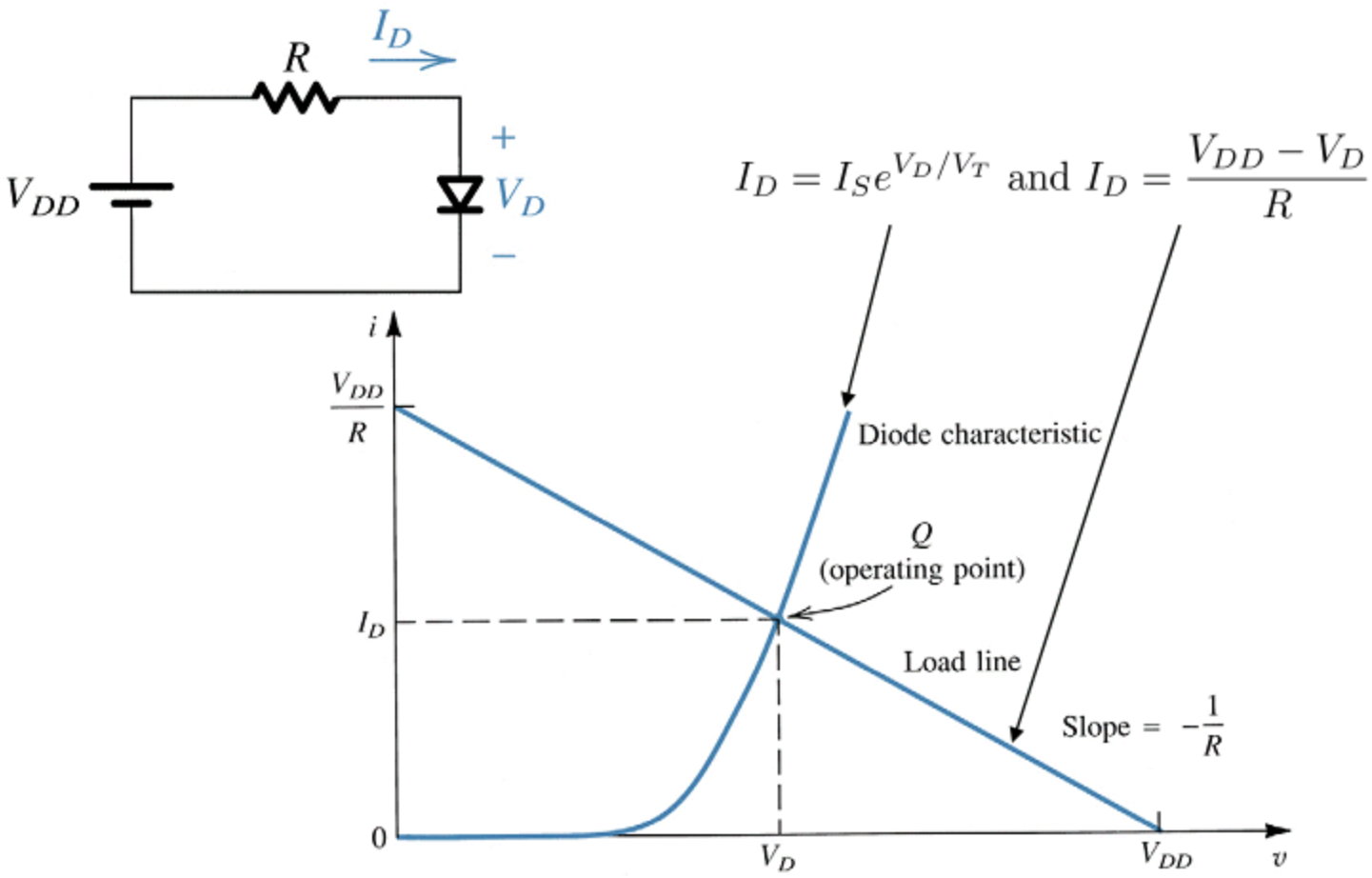
PRACTICAL Diode - multimeter



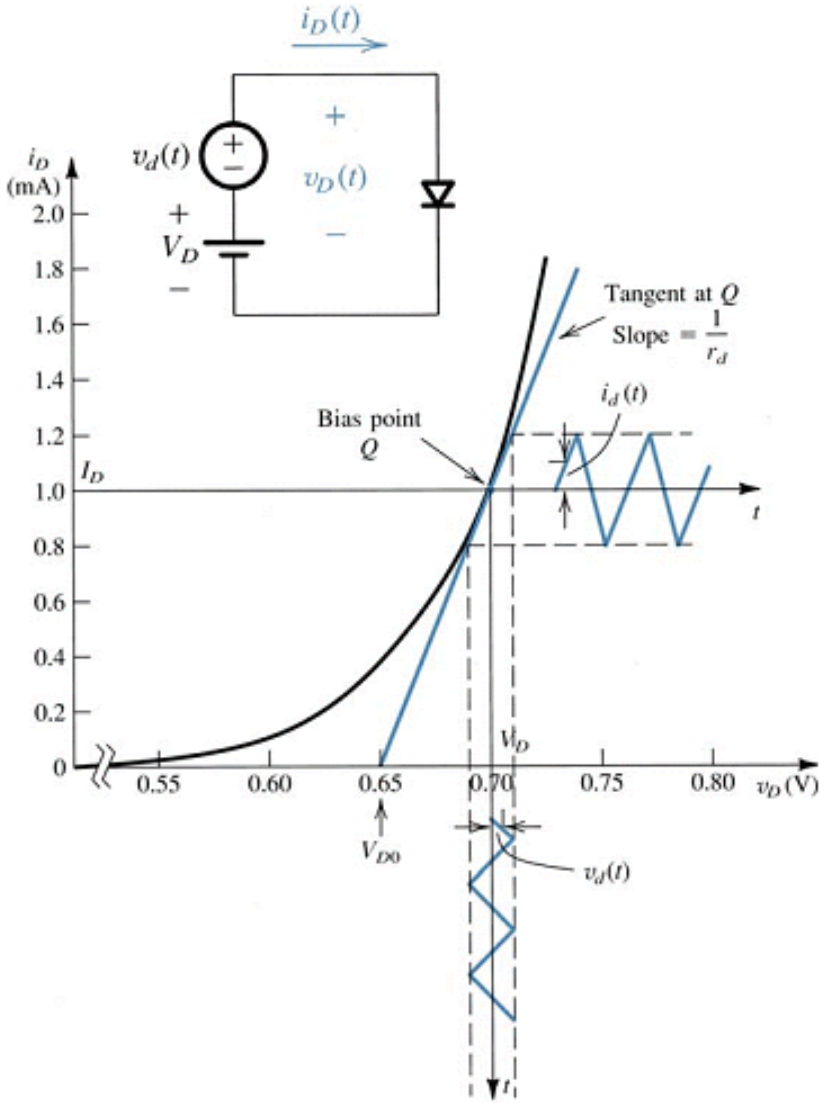
(a) Forward- and reverse-bias tests for an open diode give same indication. Some meters will display "OL."

(b) Forward- and reverse-bias tests for a shorted diode give same 0 V reading. If the diode is resistive, the reading is less than 2.6 V.

Large signal model



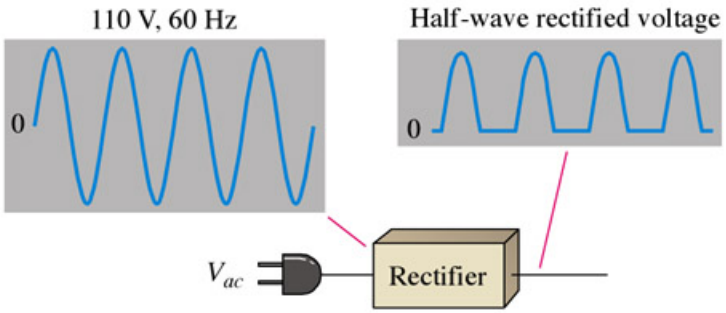
Small signal model



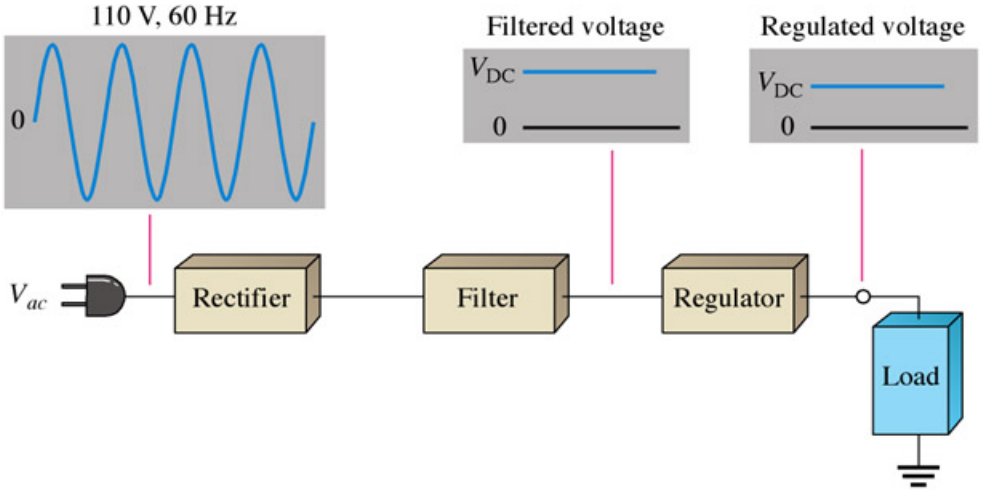
Applications

- Rectifiers
  - Half wave
  - Full wave
  - Wien bridge
  - Using capacitor (filters)
- Limiters
- Double voltage

Rectifiers

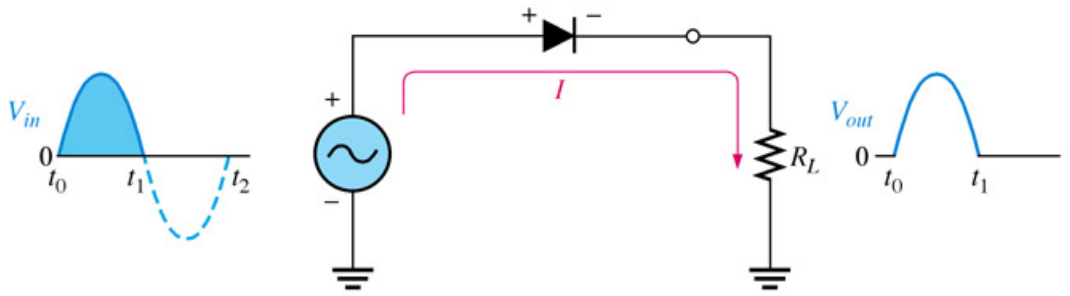


(a) Half-wave rectifier

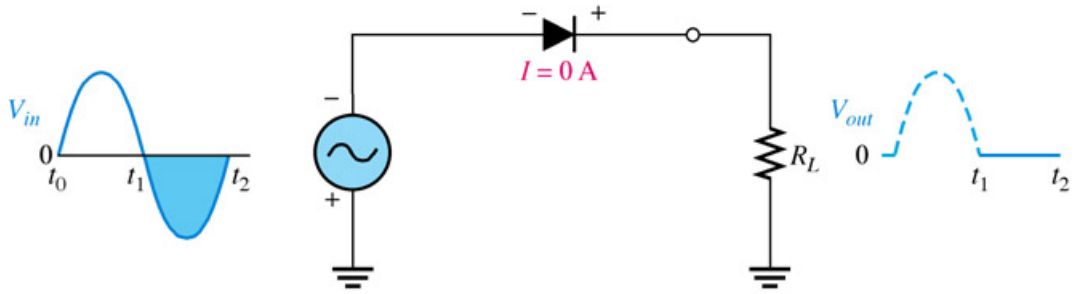


(b) Complete power supply with rectifier, filter, and regulator

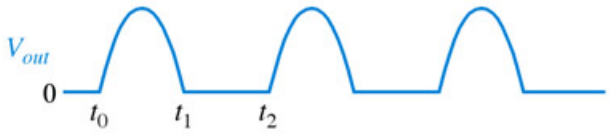
Half wave



(a) During the positive alternation of the 60 Hz input voltage, the output voltage looks like the positive half of the input voltage. The current path is through ground back to the source.



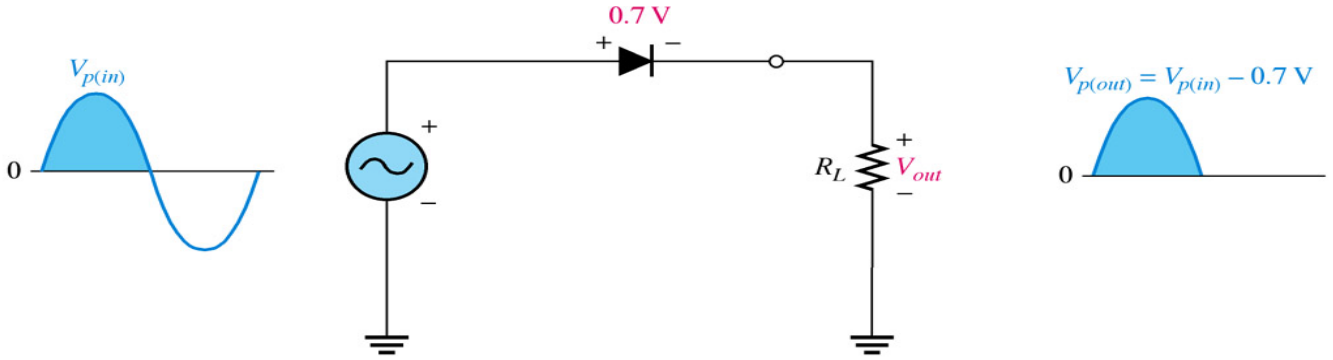
(b) During the negative alternation of the input voltage, the current is 0, so the output voltage is also 0.



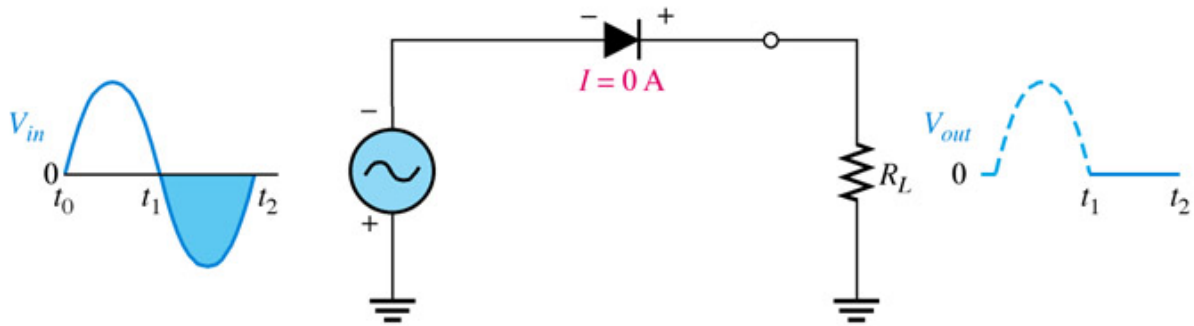
(c) 60 Hz half-wave output voltage for three input cycles

Half wave: 1st half positive periode

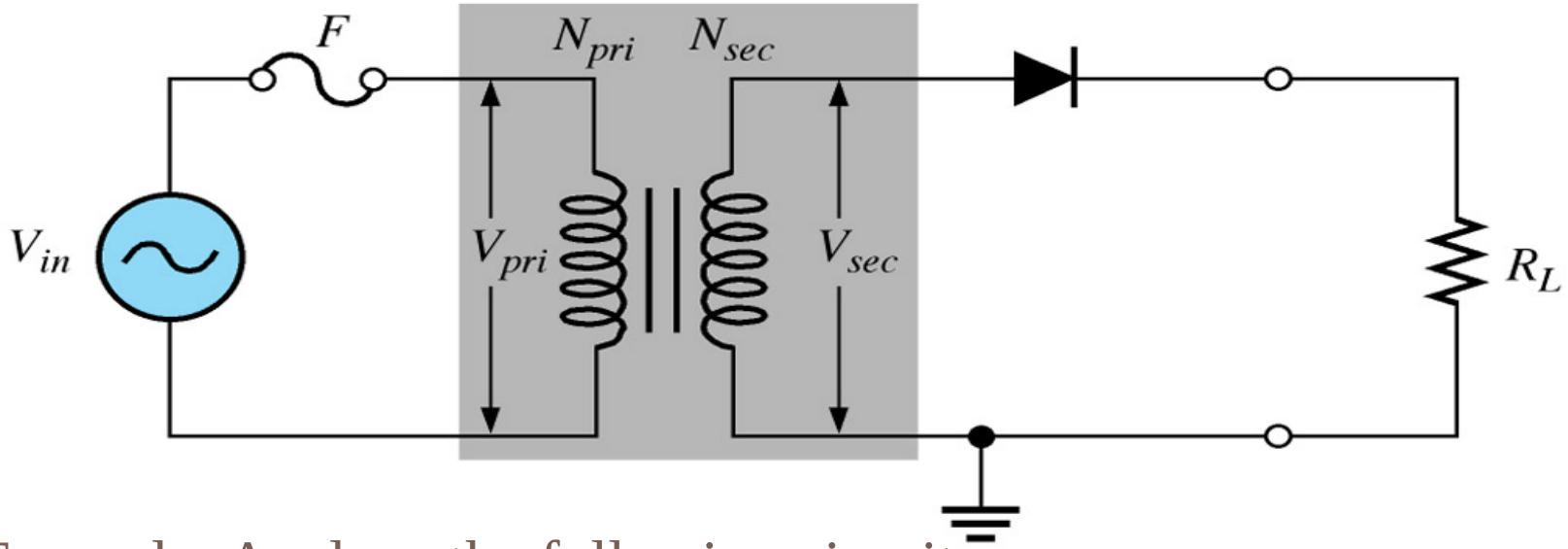
- Diode: ON
- $V_{out} = V_{in} - V_D$



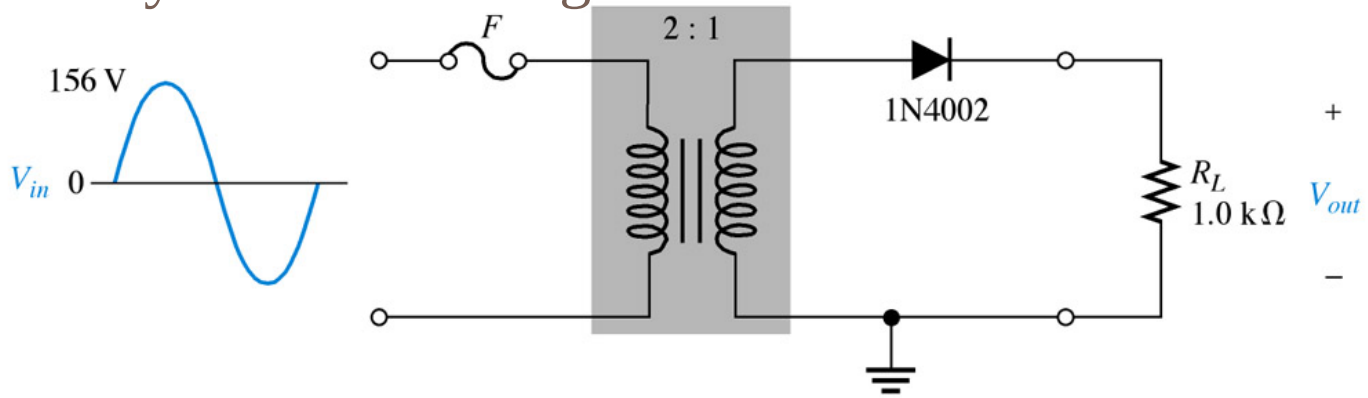
- Diode: OFF, $I_D=0$
- $V_{out} = 0$



Half wave with transformer

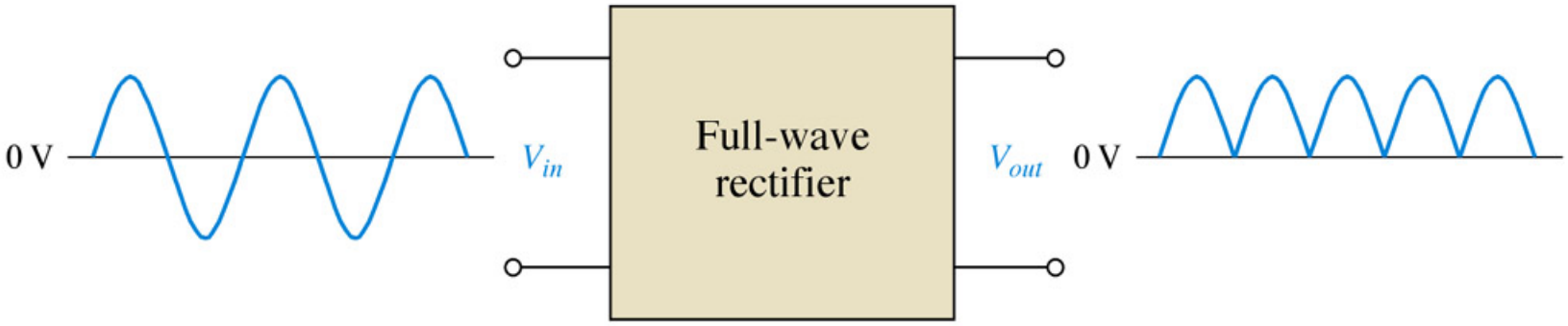


- Example: Analyze the following circuit

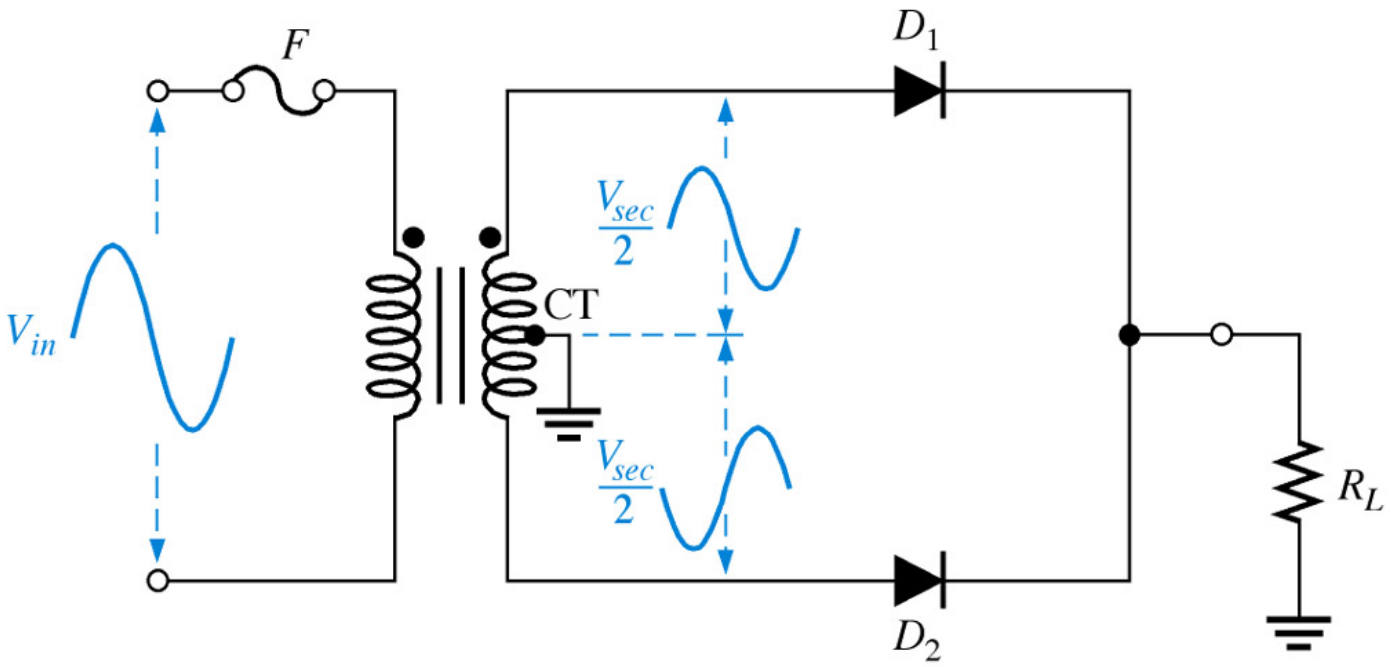


Fullwave rectifier

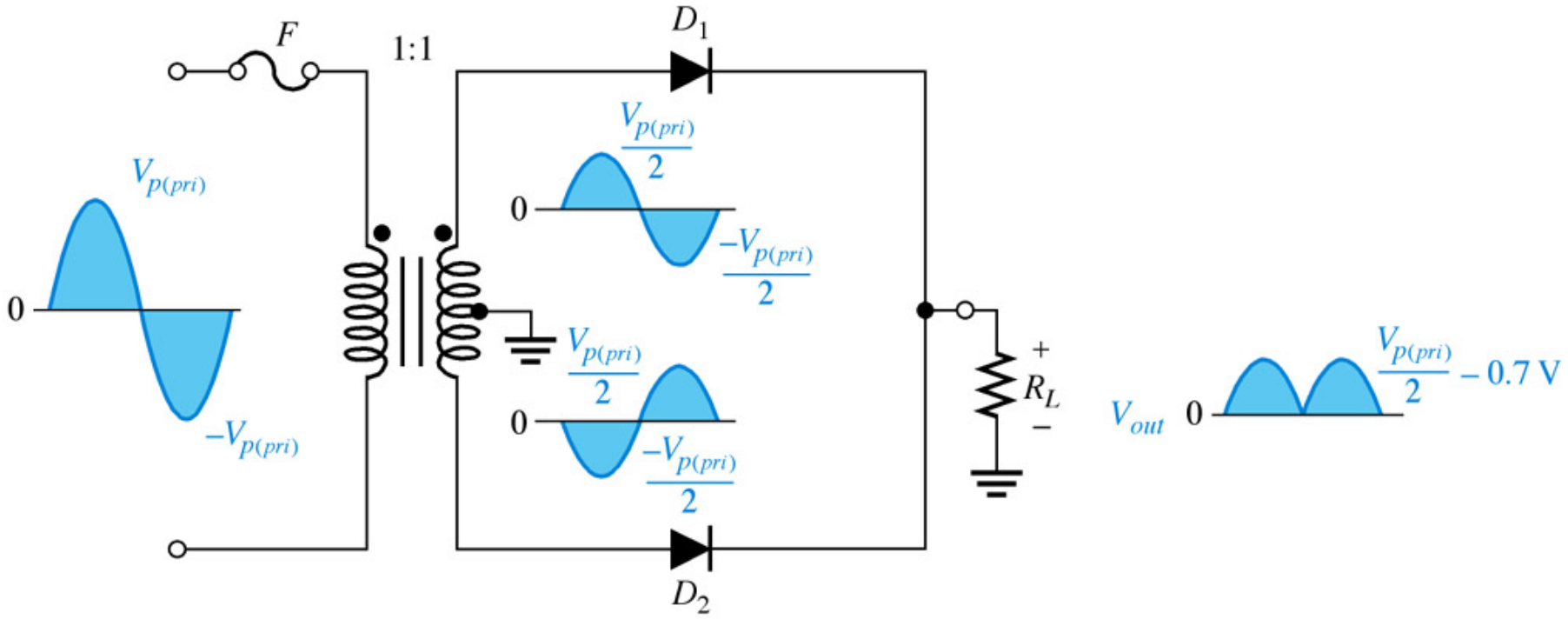
- Waveform of input and output signal



Fullwave rectifier: schematic

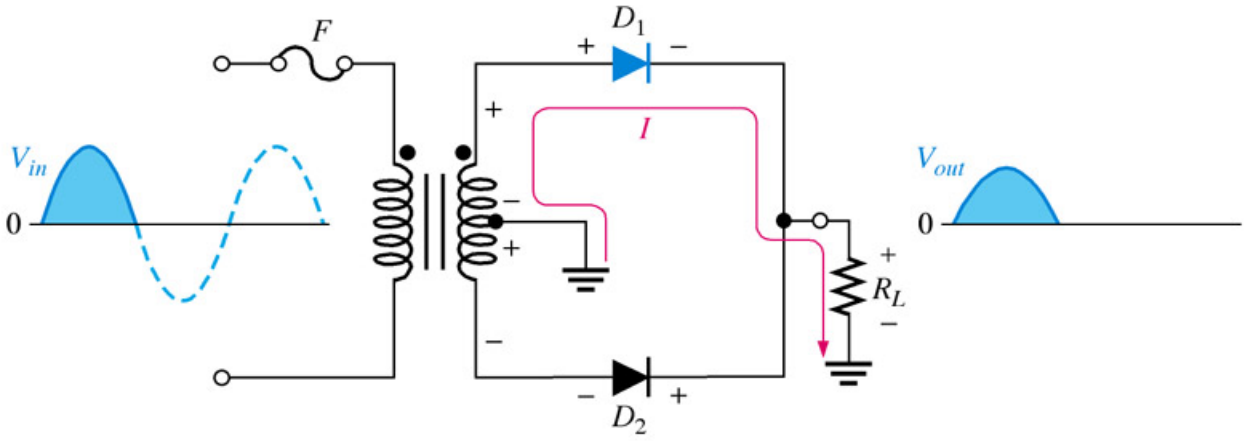


Fullwave rectifier: operation

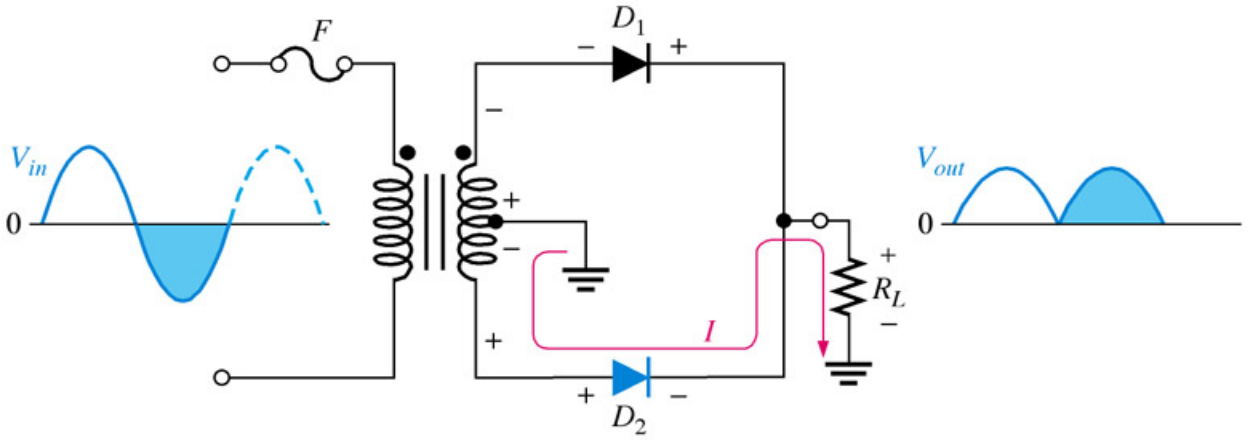


Fullwave rectifier: operation

NOTES: CURRENT flows through load resistor following a unique direction and can't be reversed



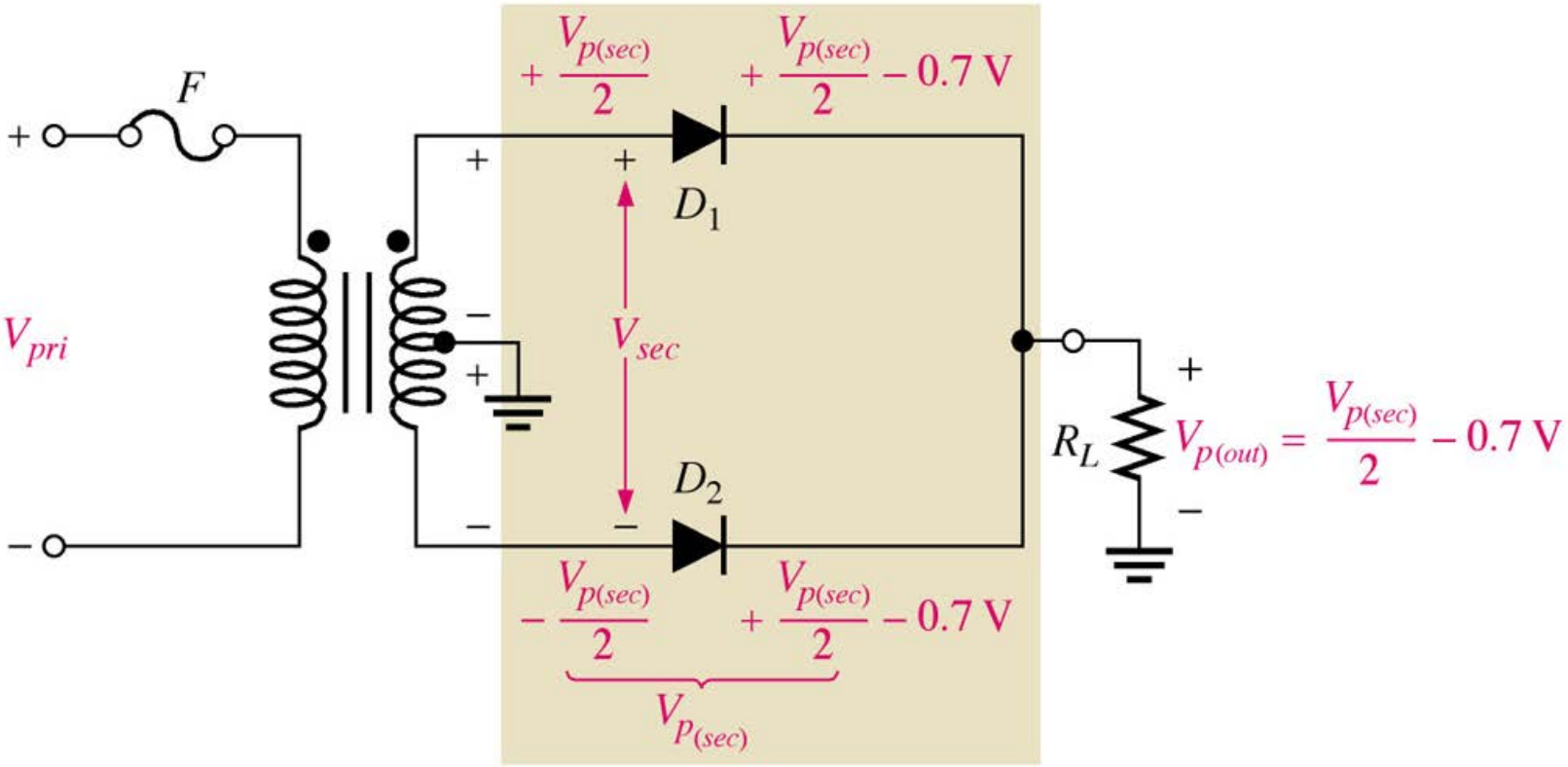
(a) During positive half-cycles, D_1 is forward-biased and D_2 is reverse-biased.



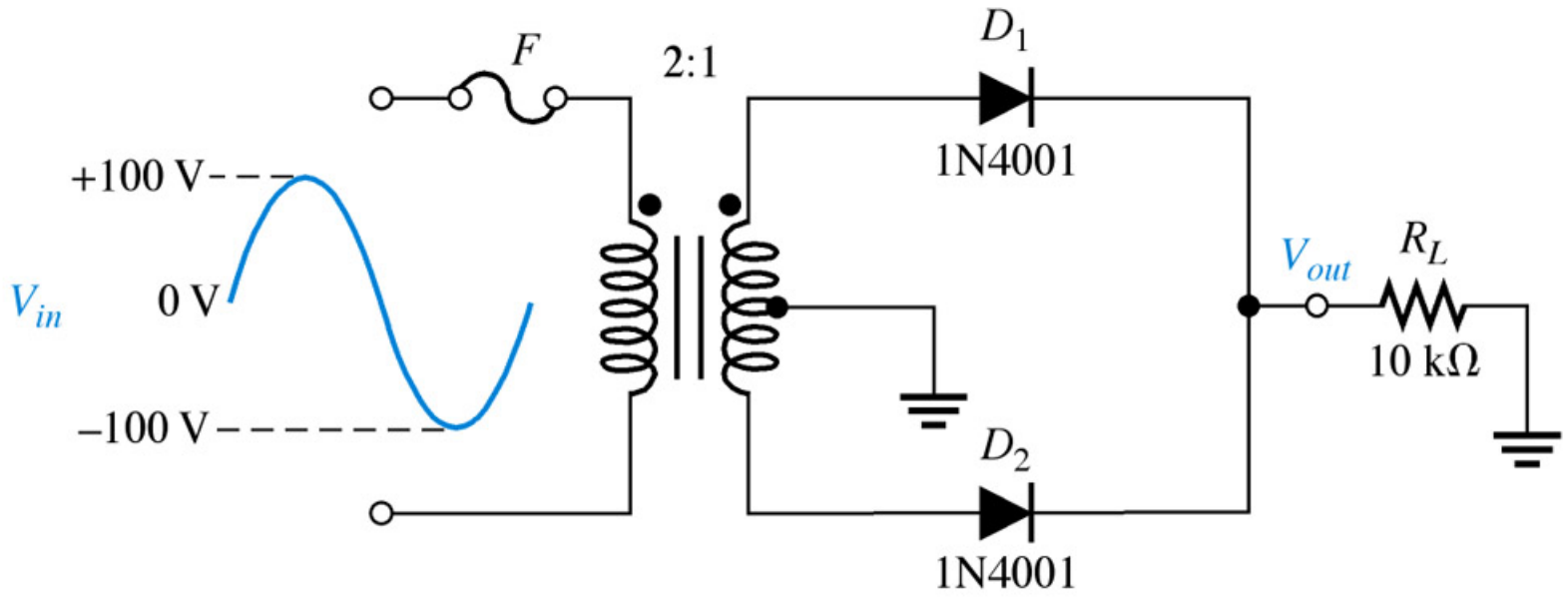
(b) During negative half-cycles, D_2 is forward-biased and D_1 is reverse-biased.

Determine PIV

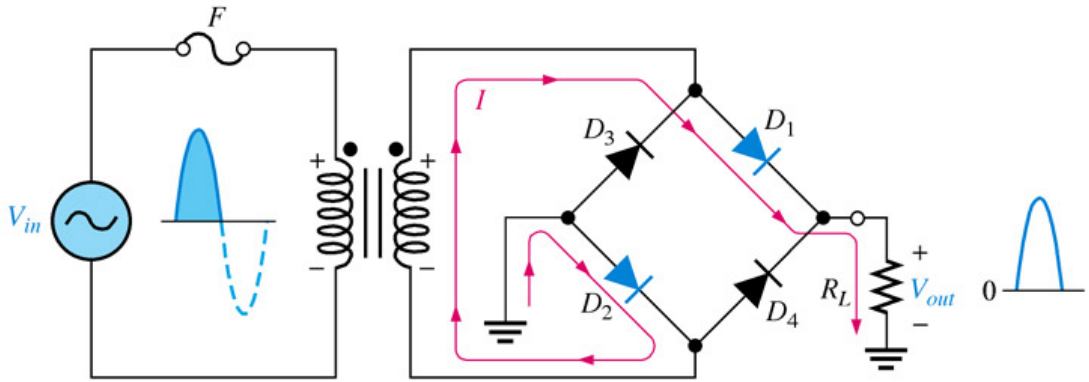
- **PIV = $V_{P(sec)}$ + 0.7V**



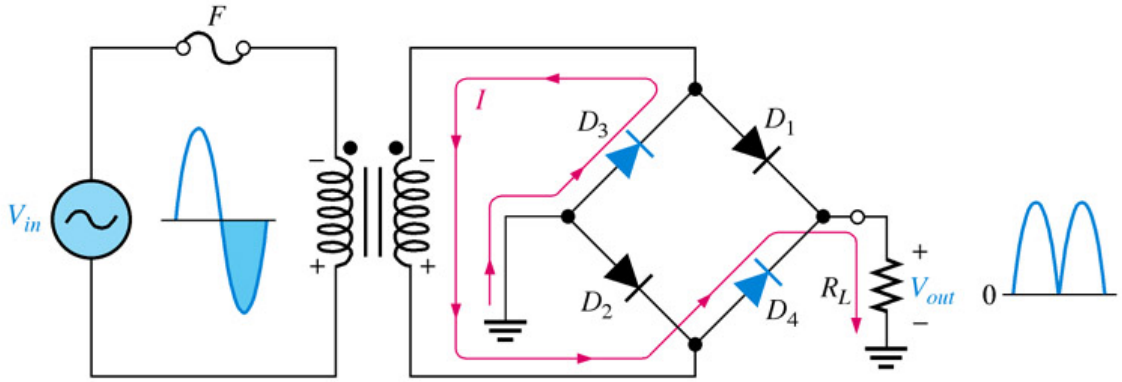
Ex: Determine PIV



Bridge rectifier

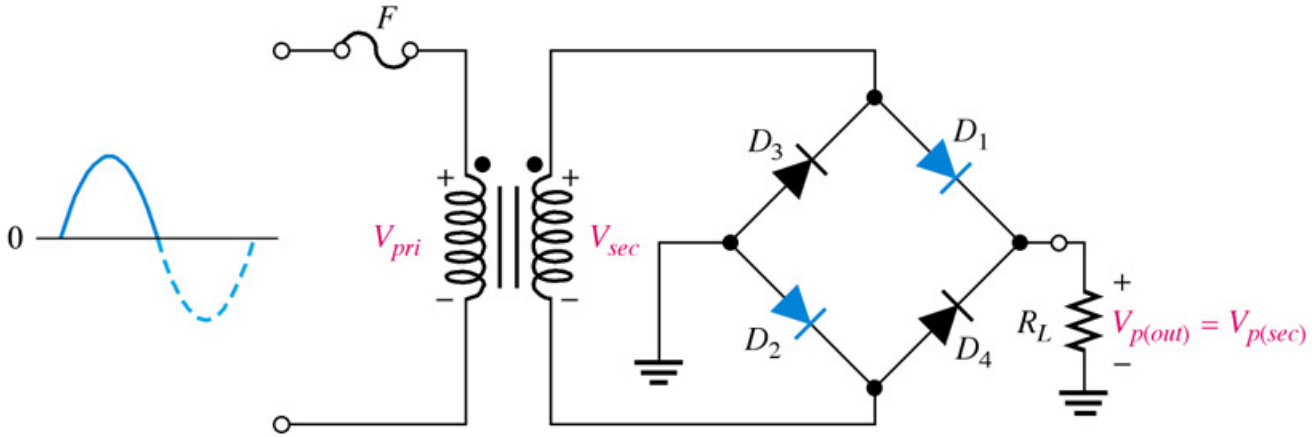


(a) During positive half-cycle of the input, D_1 and D_2 are forward-biased and conduct current. D_3 and D_4 are reverse-biased.

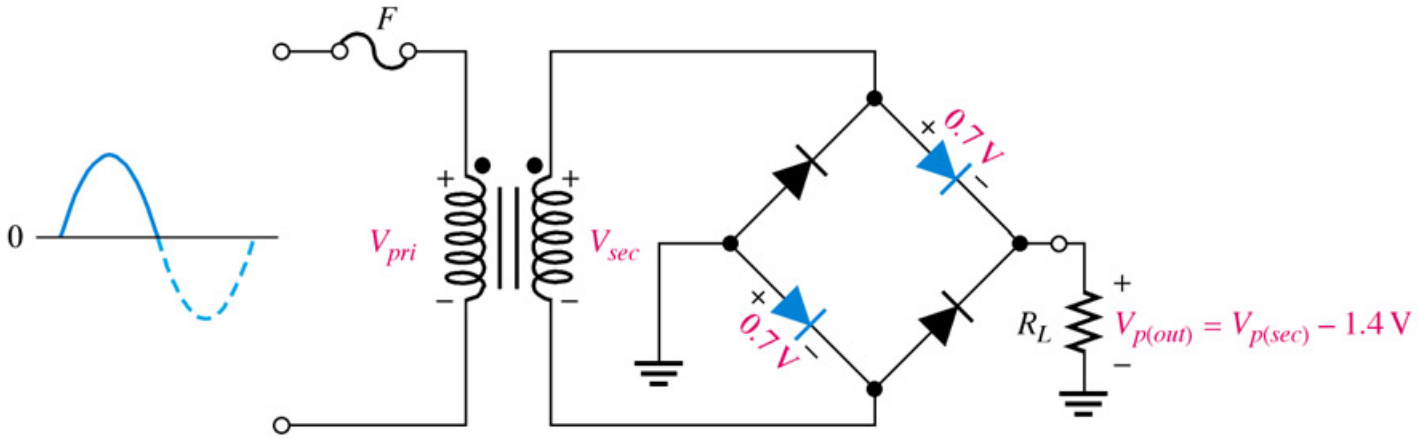


(b) During negative half-cycle of the input, D_3 and D_4 are forward-biased and conduct current. D_1 and D_2 are reverse-biased.

Bridge rectifier



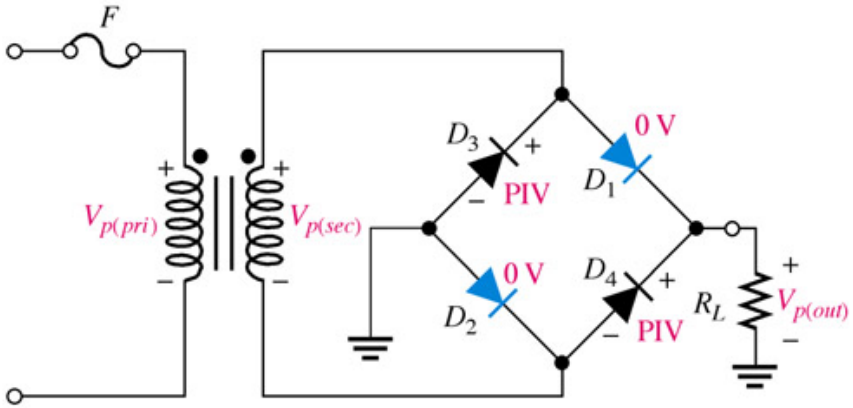
(a) Ideal diodes



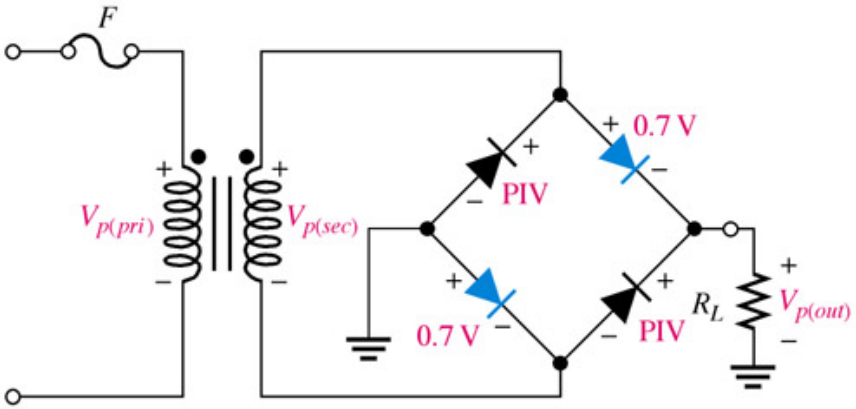
(b) Practical diodes (Diode drops included)

Bridge rectifier

PIV = $V_{P(OUT)}$ + 0.7V

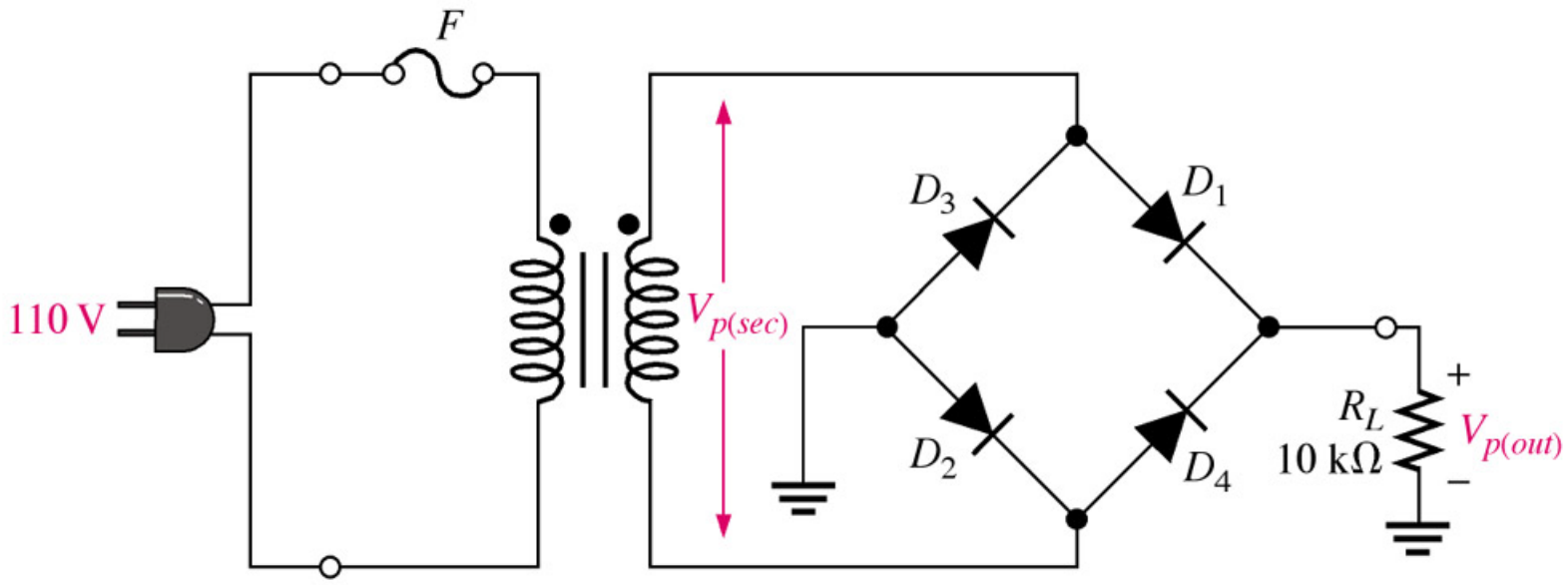


(a) For the ideal diode model (forward-biased diodes D_1 and D_2 are shown in blue). PIV = $V_{p(out)}$

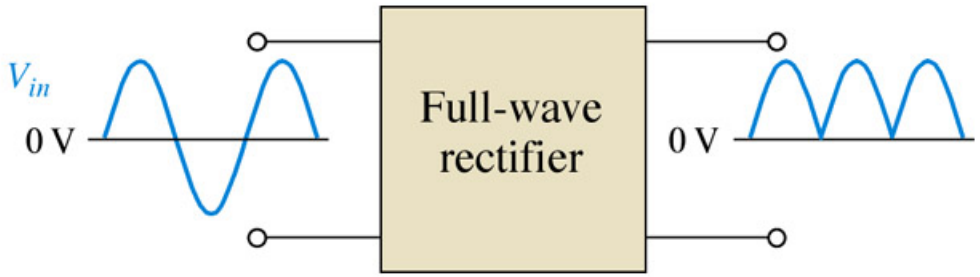


(b) For the practical diode model (forward-biased diodes D_1 and D_2 are shown in blue). PIV = $V_{p(out)}$ + 0.7 V

Bridge rectifier



Filters

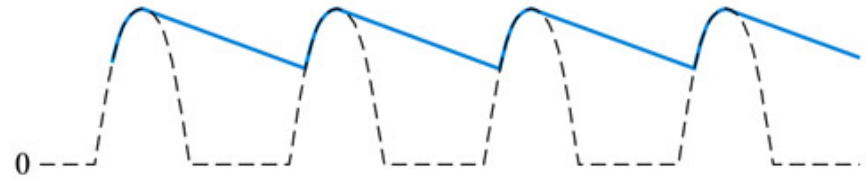


(a) Rectifier without a filter

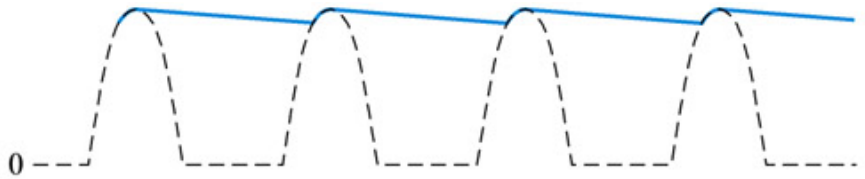


(b) Rectifier with a filter (output ripple is exaggerated)

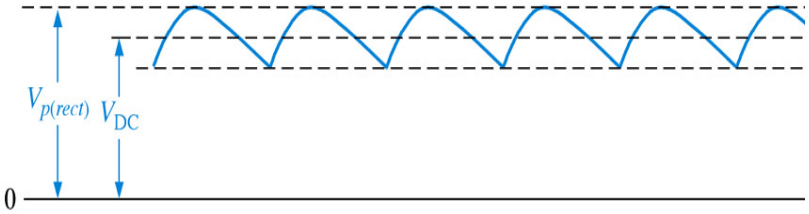
Roles of Capacitor

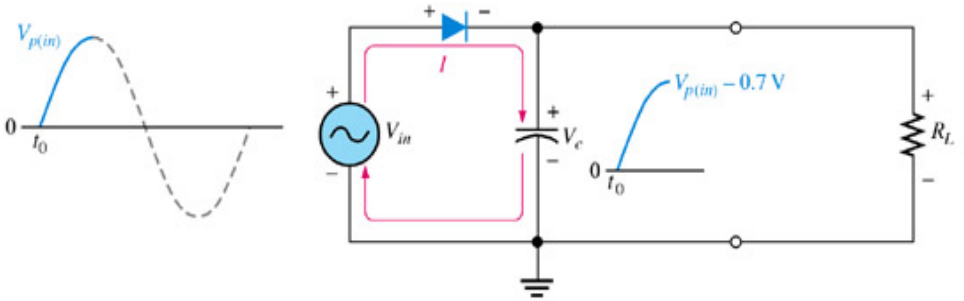


(a) Larger ripple means less effective filtering.

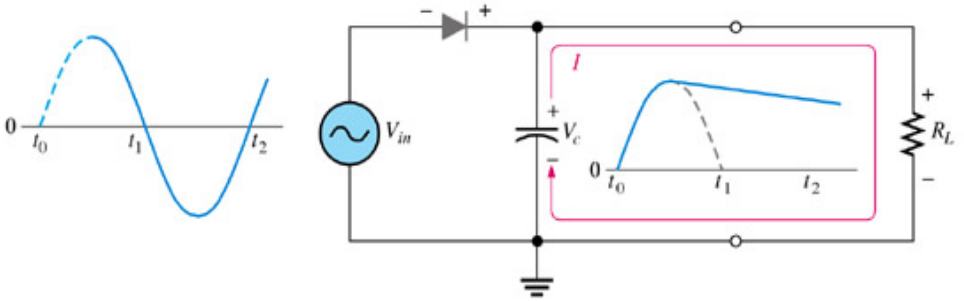


(b) Smaller ripple means more effective filtering. Generally, the larger the capacitor value, the smaller the ripple for the same input and load.

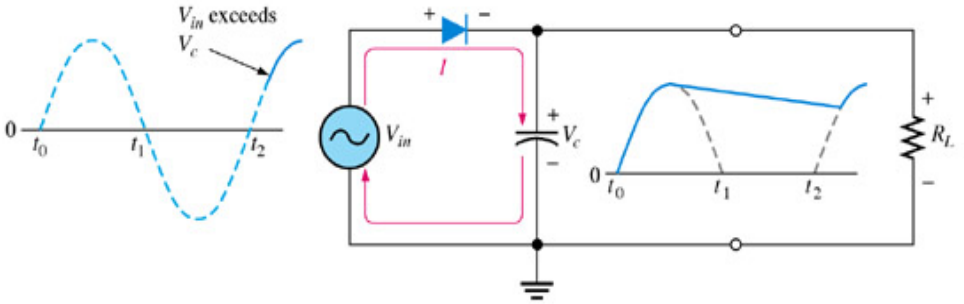




(a) Initial charging of capacitor (diode is forward-biased) happens only once when power is turned on.



(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid blue curve.



(c) The capacitor charges back to peak of input when the diode becomes forward-biased. This charging occurs during the portion of the input voltage indicated by the solid blue curve.

Output frequency

- Input: Sinusoidal waveform, imput f: 60Hz



$$f_{hw} = \frac{1}{T_{hw}} = \frac{1}{16.7\ \text{ms}} = 60\ \text{Hz}$$

(a) Half-wave (*hw*)



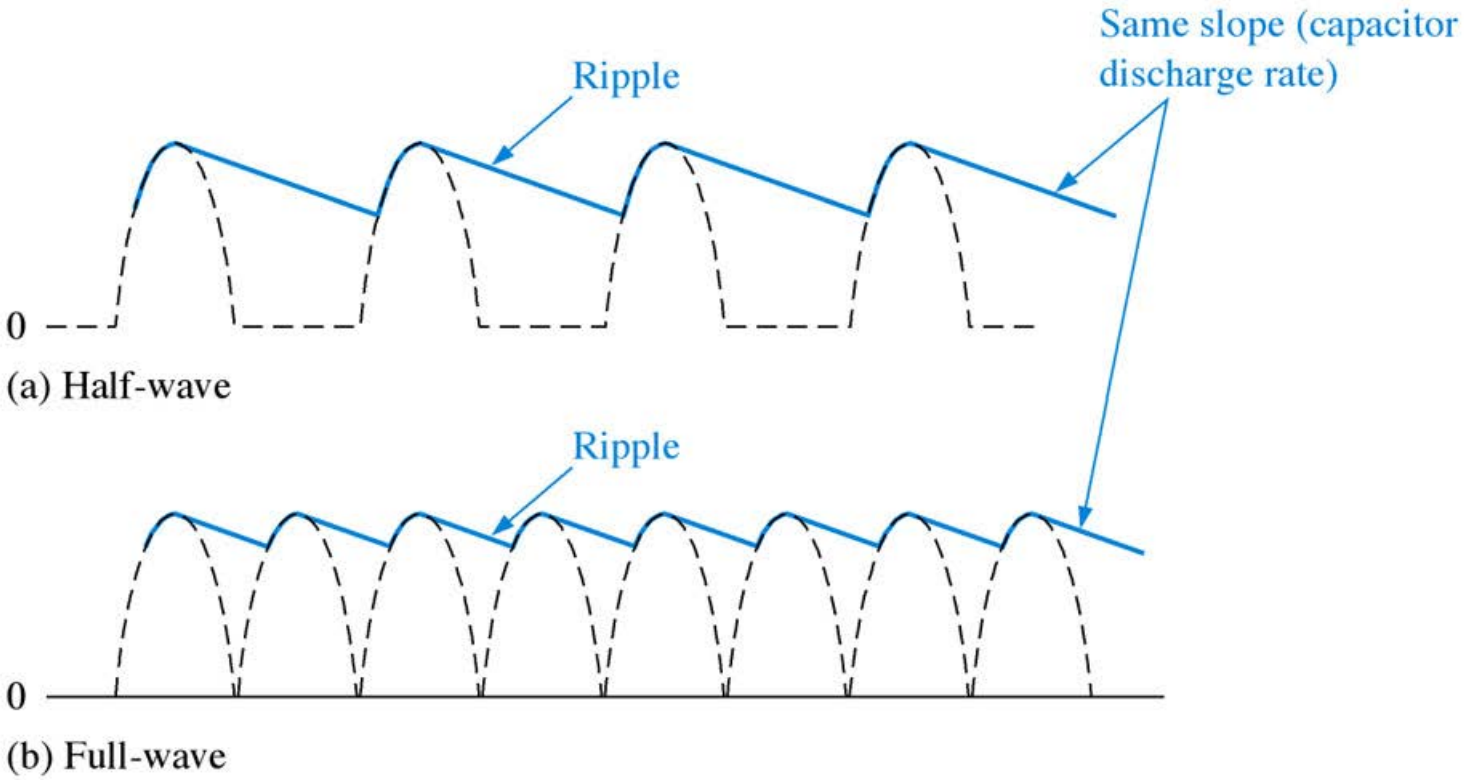
$$T_{fw} = \frac{T_{hw}}{2}$$

$$f_{fw} = \frac{1}{T_{fw}} = 2\left(\frac{1}{T_{hw}}\right) = 2\,f_{hw}$$

$$= 2(60\ \text{Hz}) = 120\ \text{Hz}$$

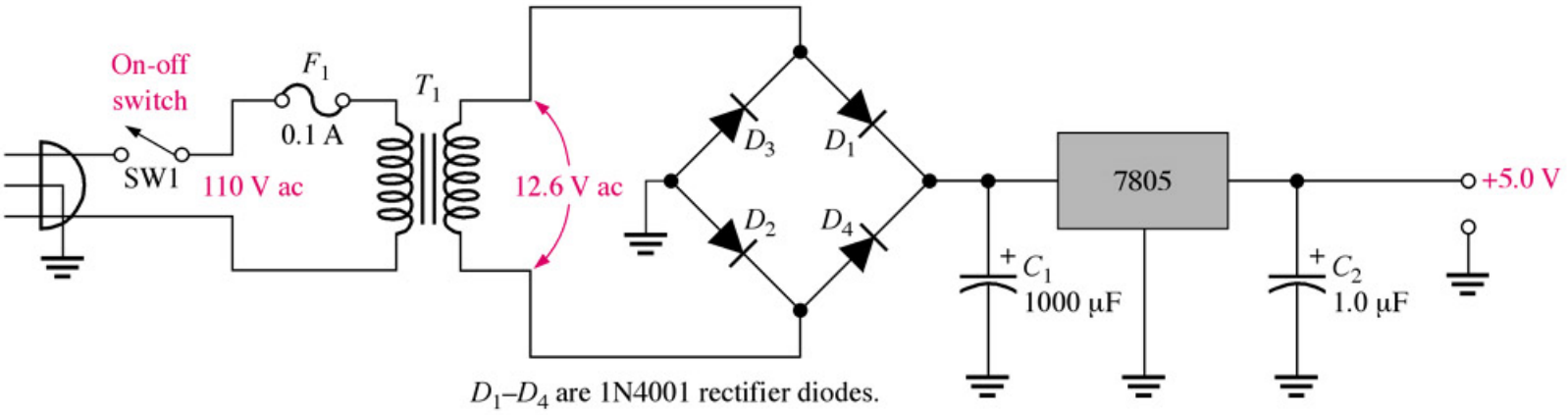
Ripple consideration

- Ripples observed on output signals
  Same filter cap, load and input signal



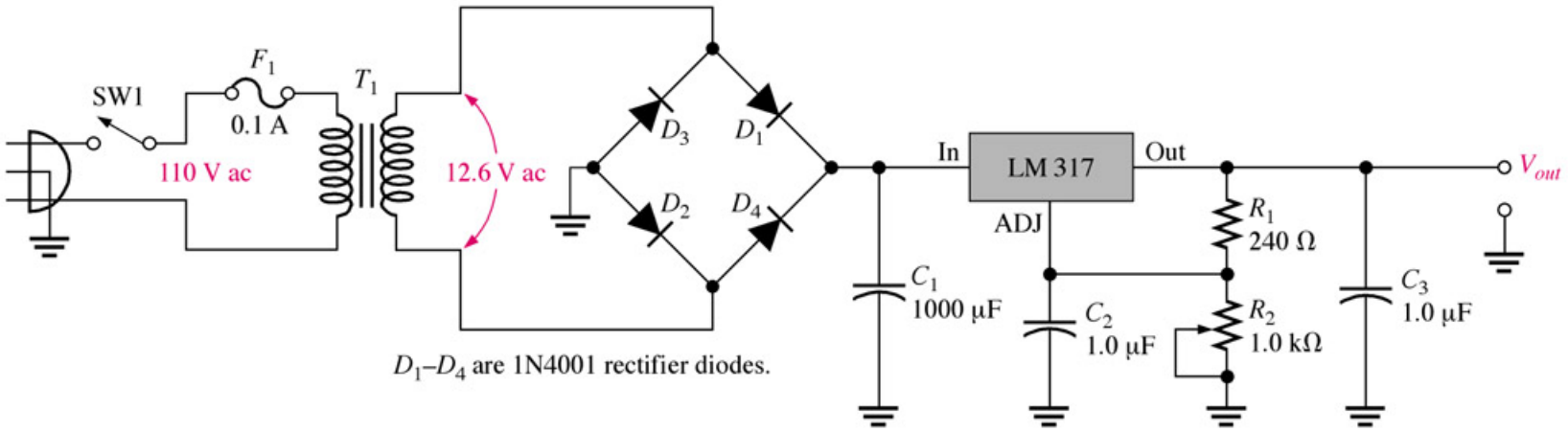
A supply voltage

- Used in most electronics devices
- Simple, cheap and easy to construct

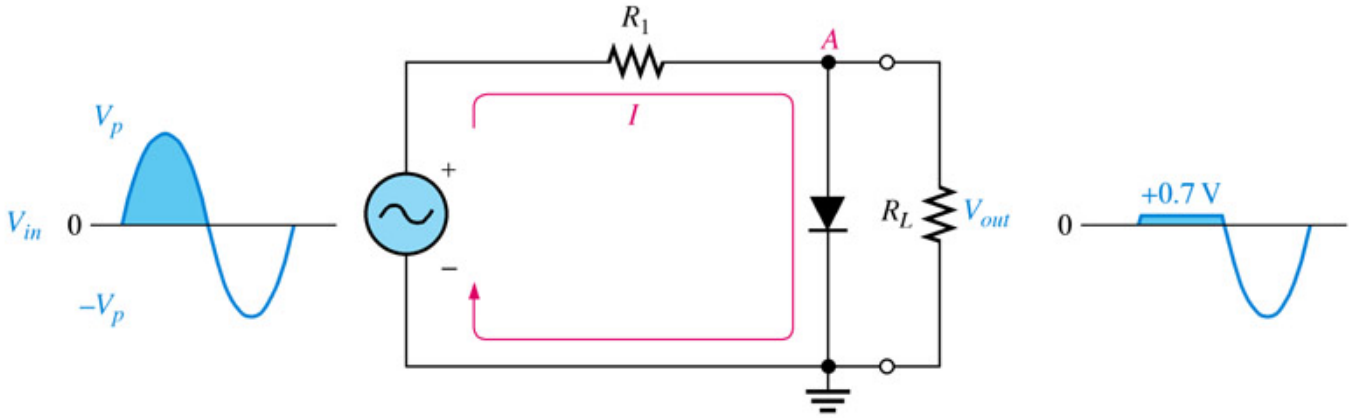


A supply voltage –

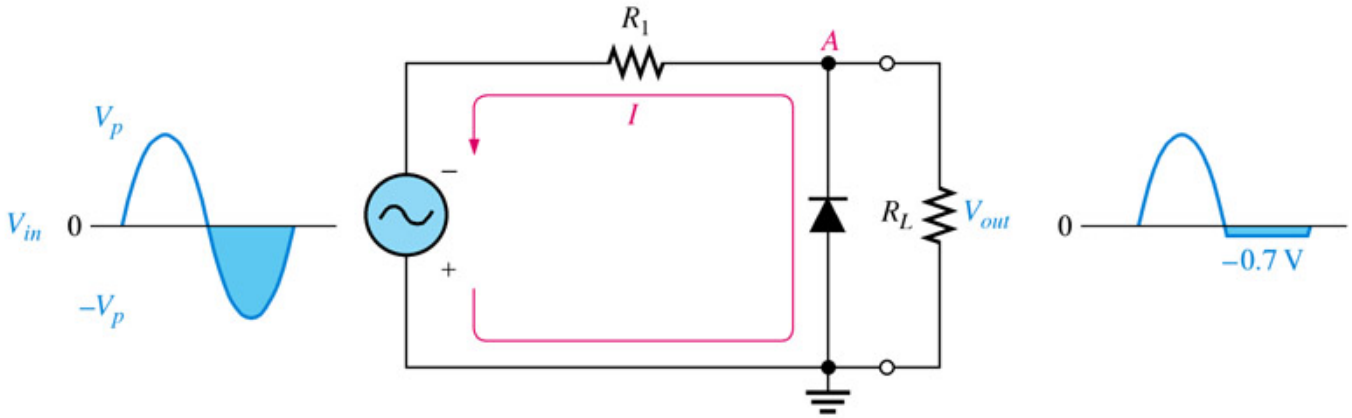
- output from 1.25 to 6.5V



Limiters



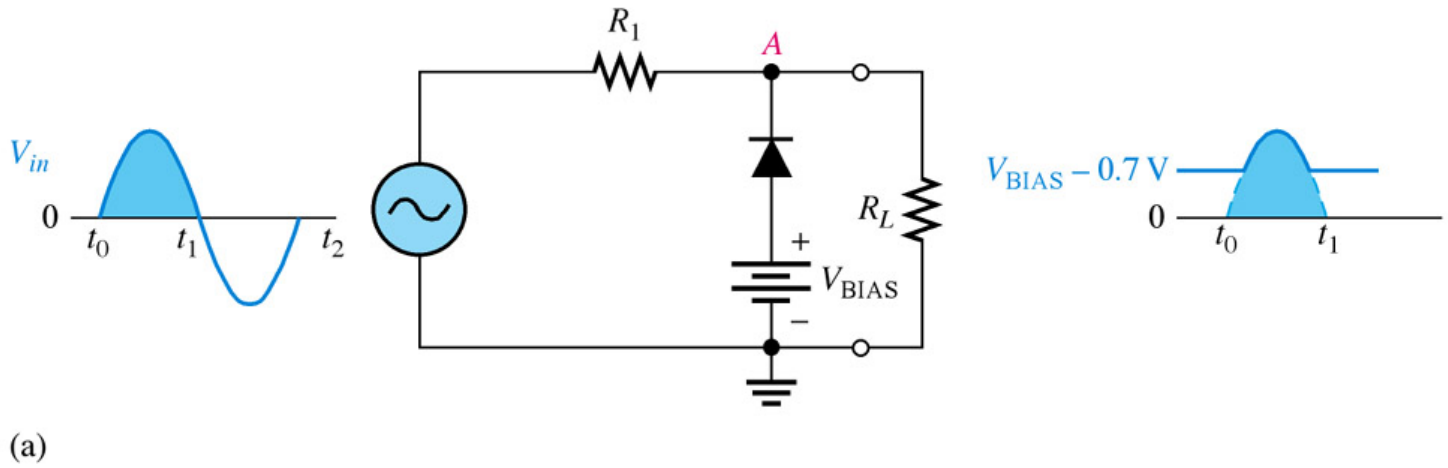
(a) Limiting of the positive alternation. The diode is forward-biased during the positive alternation (above 0.7 V) and reverse-biased during the negative alternation.



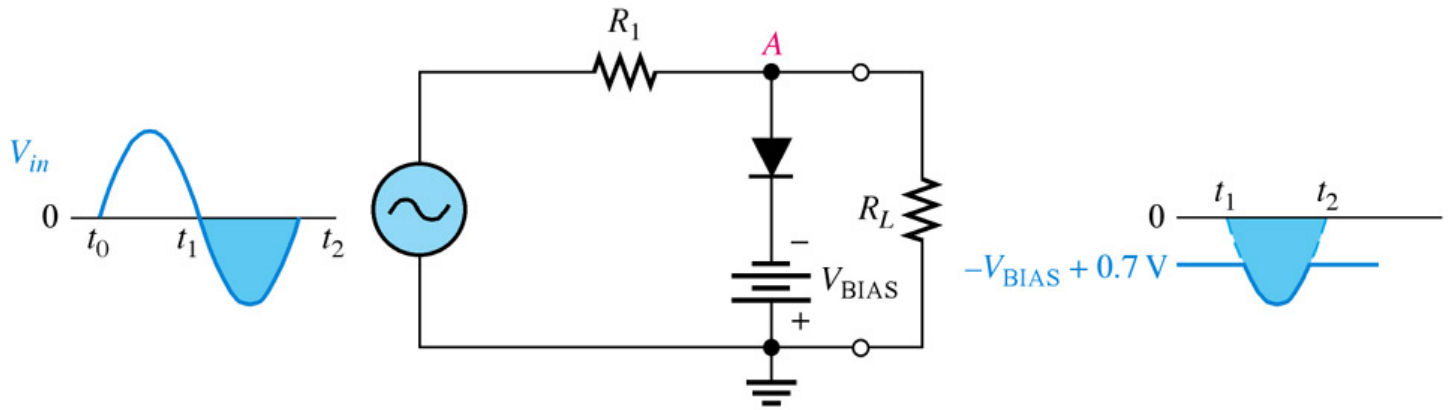
(b) Limiting of the negative alternation. The diode is forward-biased during the negative alternation (below −0.7 V) and reverse-biased during the positive alternation.

Different operation of diodes

- Input and output waveforms



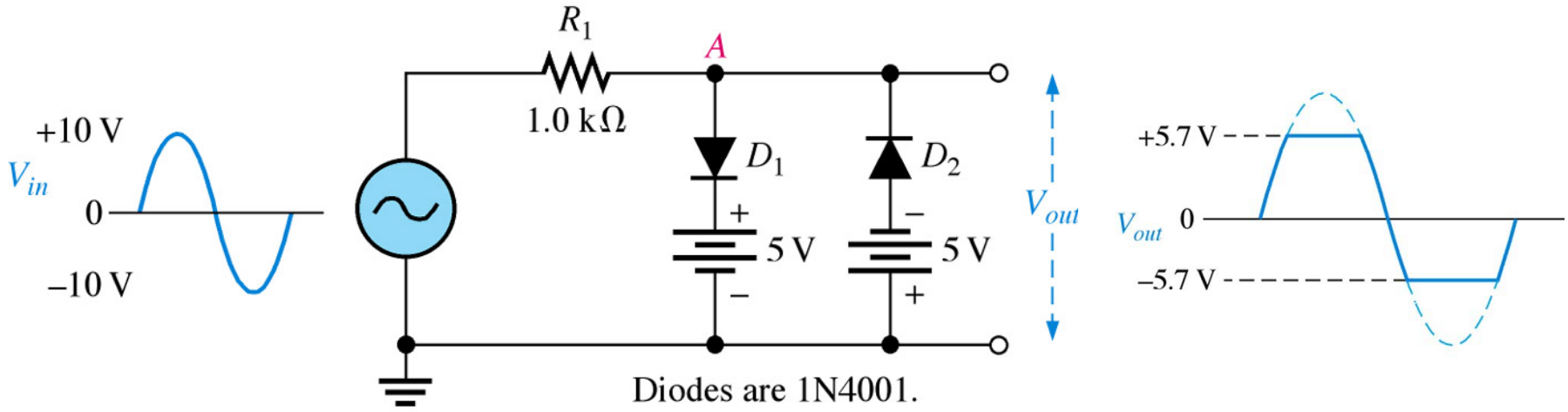
(a)



(b)

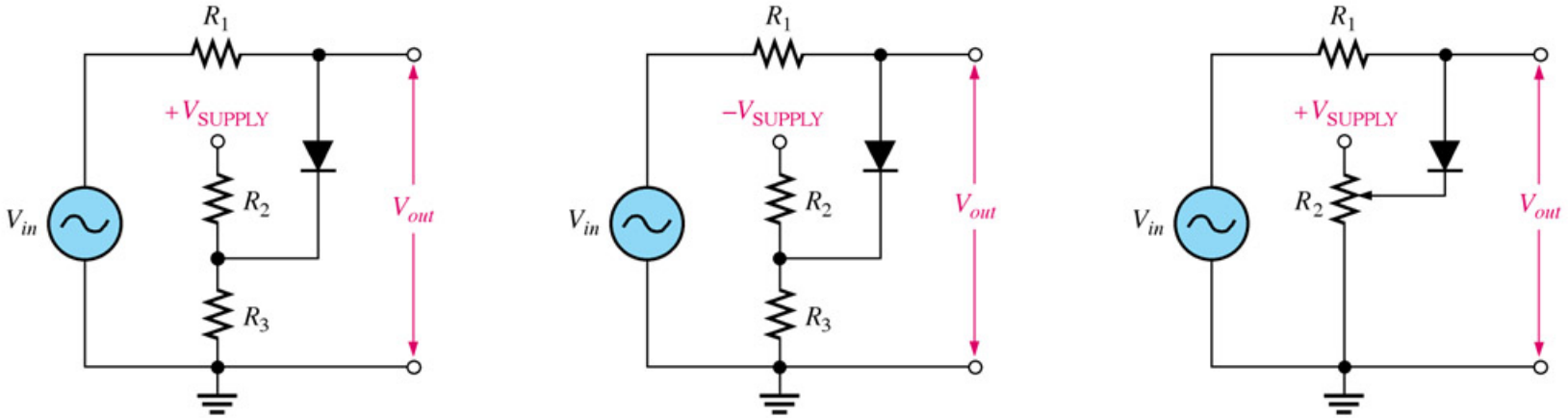
Similar Zener diode operation

- Input and output waveforms



Limiters associated with voltage divider

Ex: Analyze this circuit



(a) Positive limiter

(b) Negative limiter

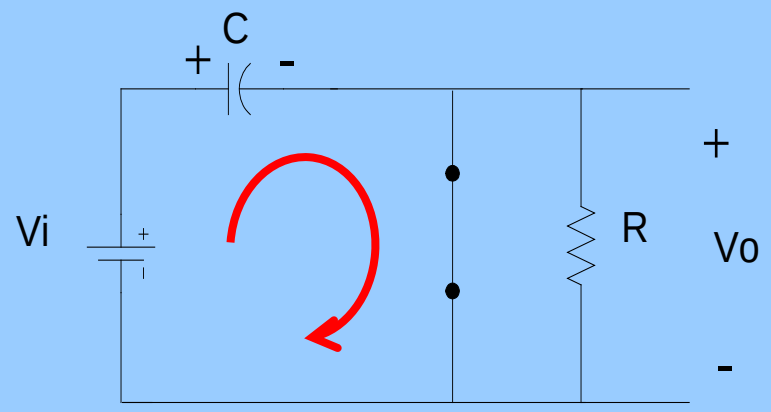
(c) Variable positive limiter

Clamp circuit

- Displace the DC level at output
- Consisting of:
  - ✓ Capacitor
  - ✓ Diode
  - ✓ Resistors
  - ✓ battery
- Resistor and capacitor are chosen how to guarantee a τ = RC big enough so that the discharge time of capacitor is still higher than the reverse period of diode

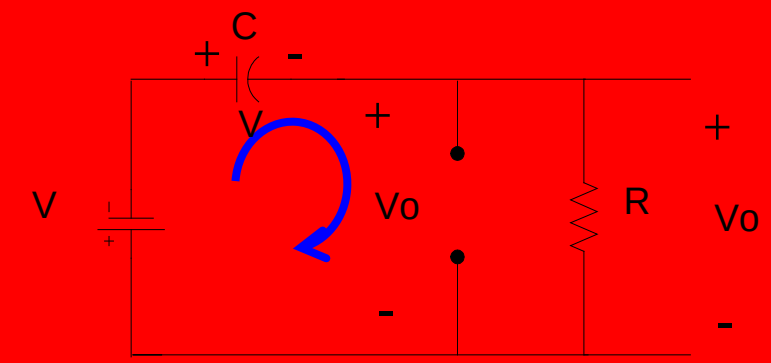
Clamp operation

1st half period (+)



- 0→ T/2: Diode ON (short circuit)
- C charges to level V_i
- V_o= 0 V

2nd half period (-)



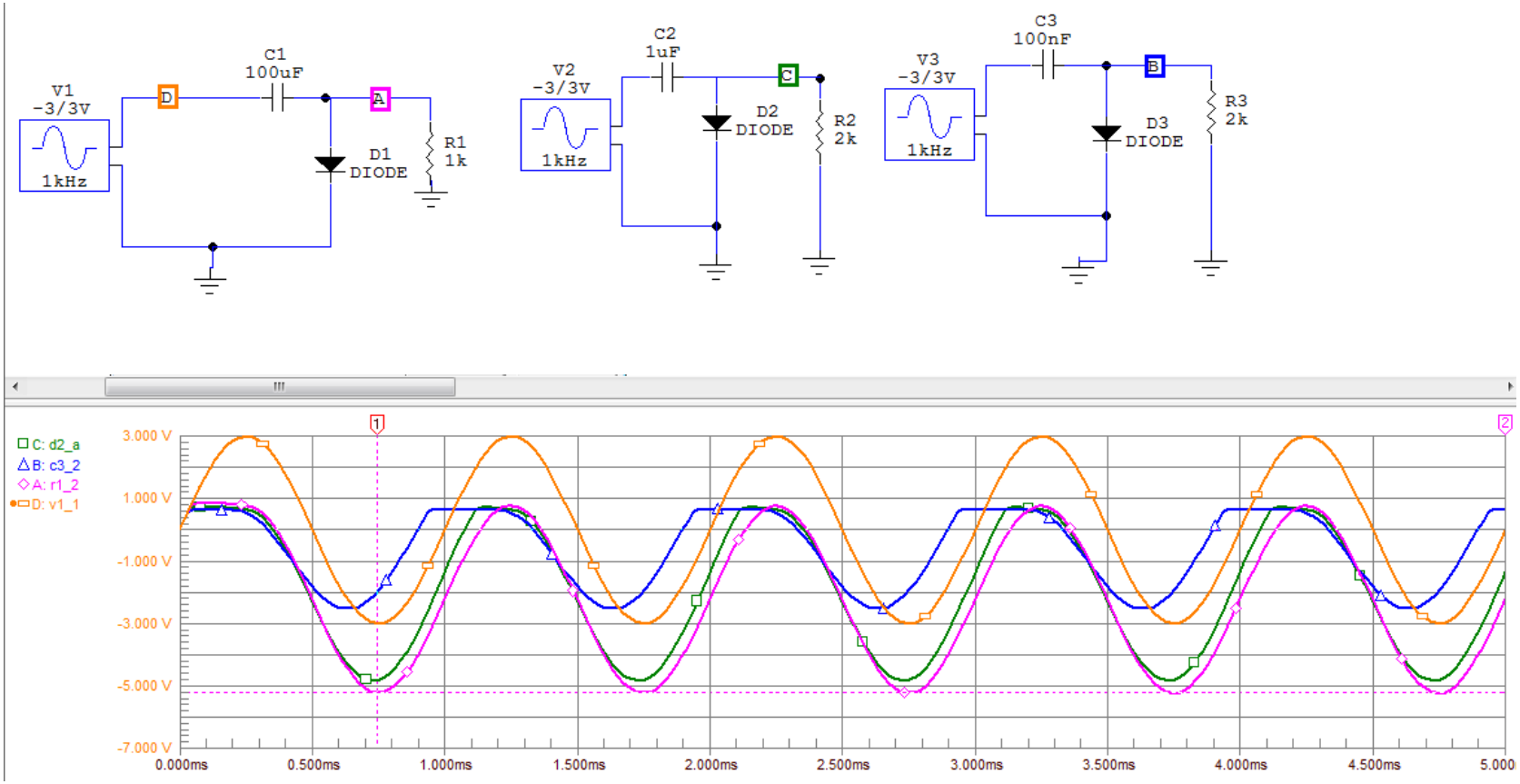
- T/2→ T: Diode OFF (open circuit),

Kirchoff Voltage Law:

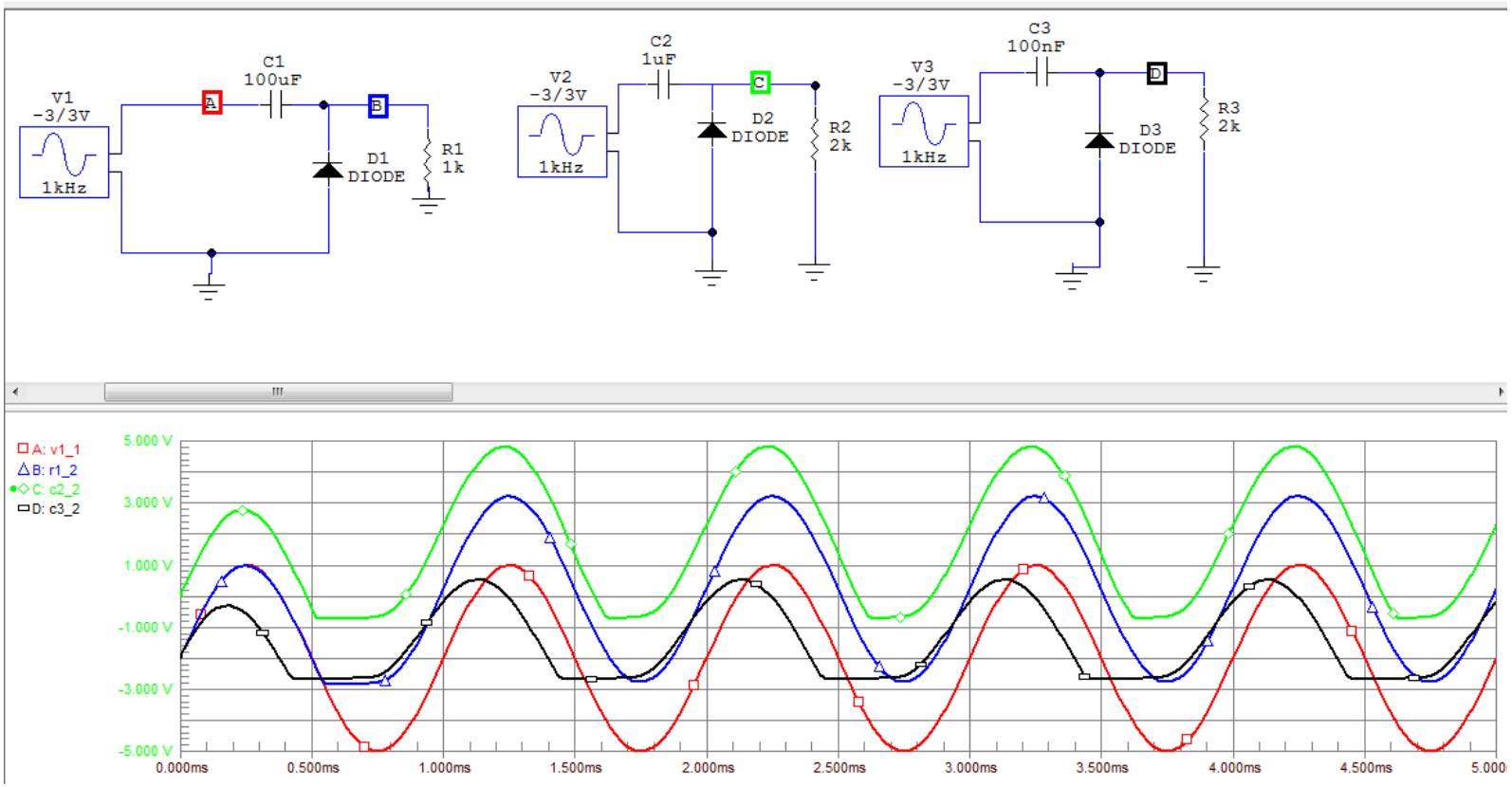
$-V - V - V_o = 0$ and

➔ $V_o = -2V$

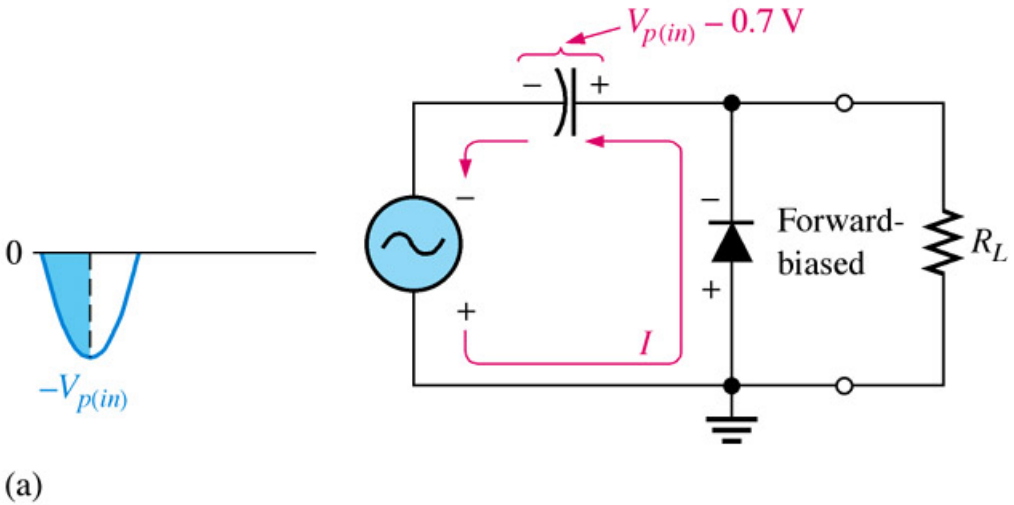
Results from Circuit Maker



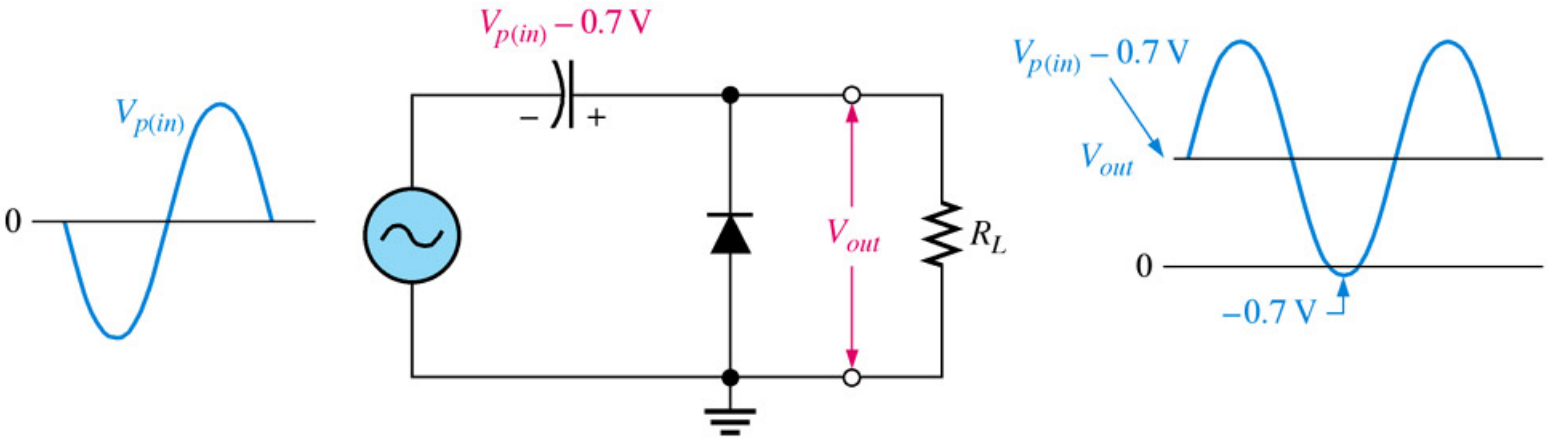
Results from Circuit Maker



Positive Clamp circuit

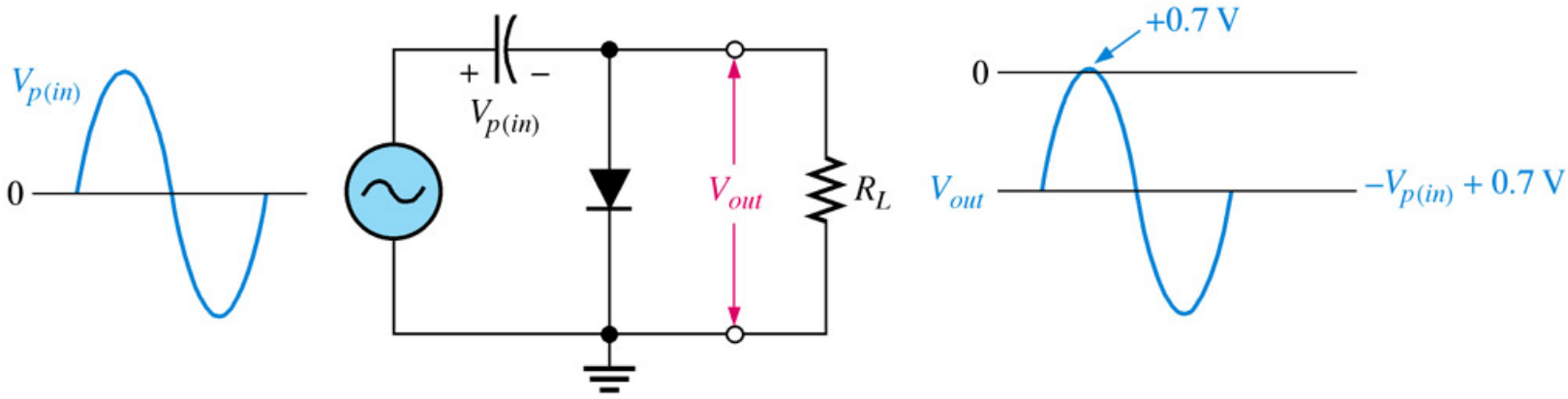


(a)

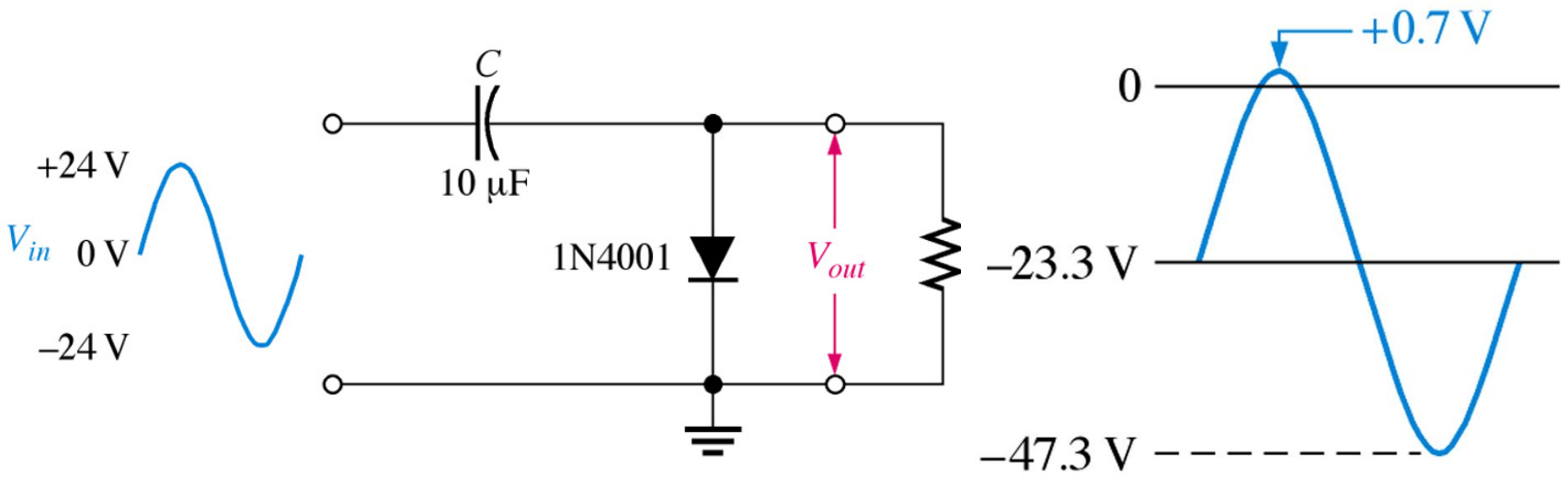


(b)

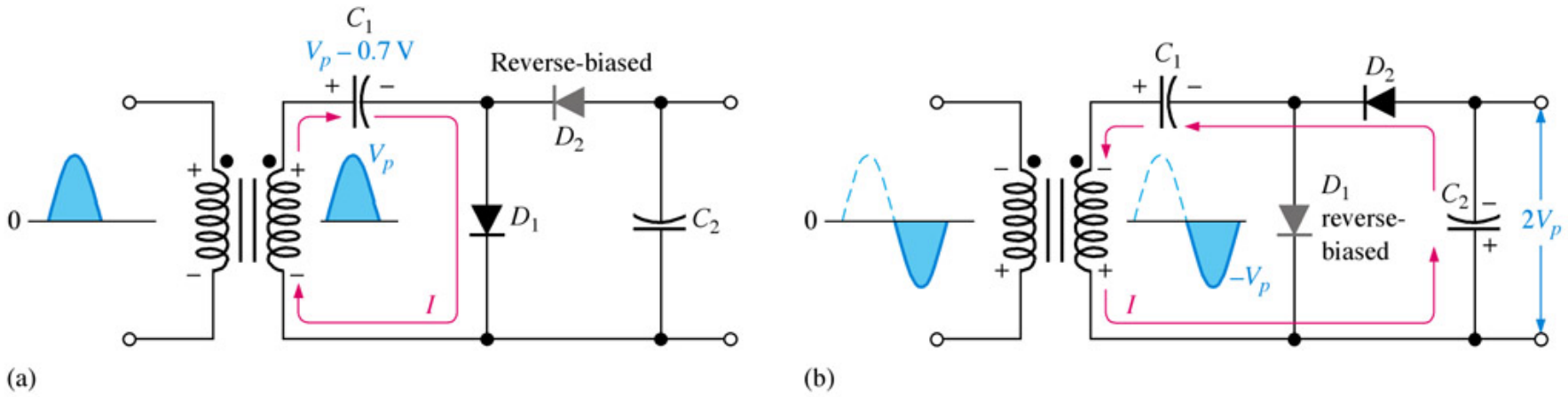
Negative clamp circuit



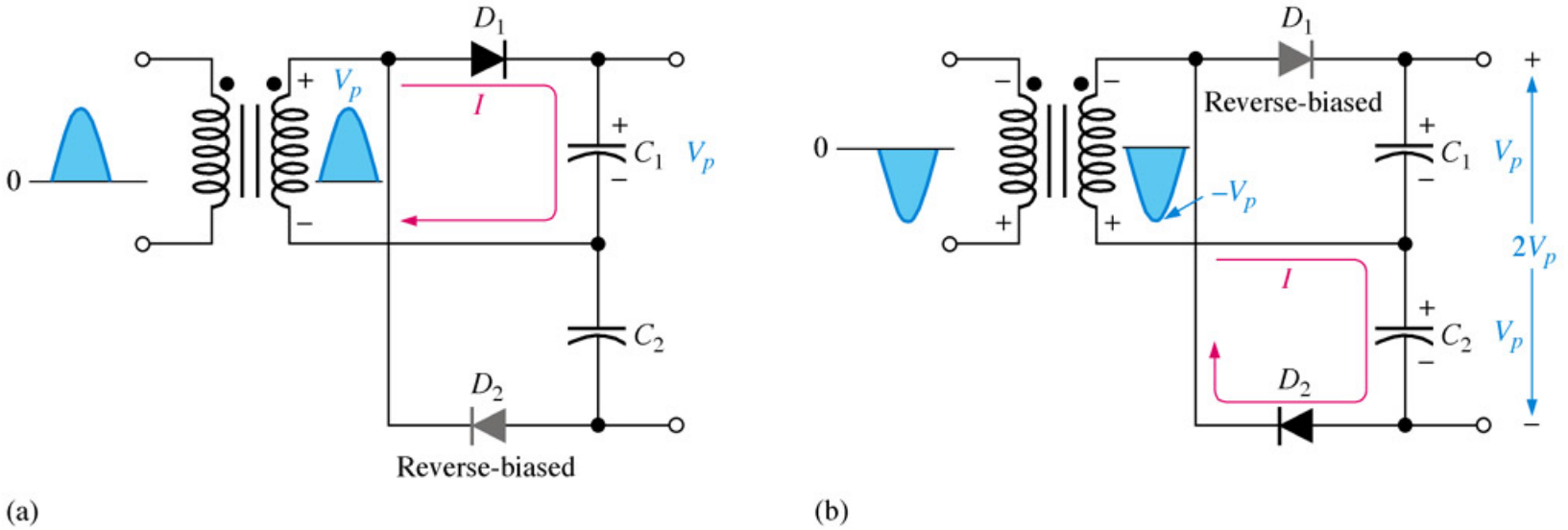
Negative clamp circuit



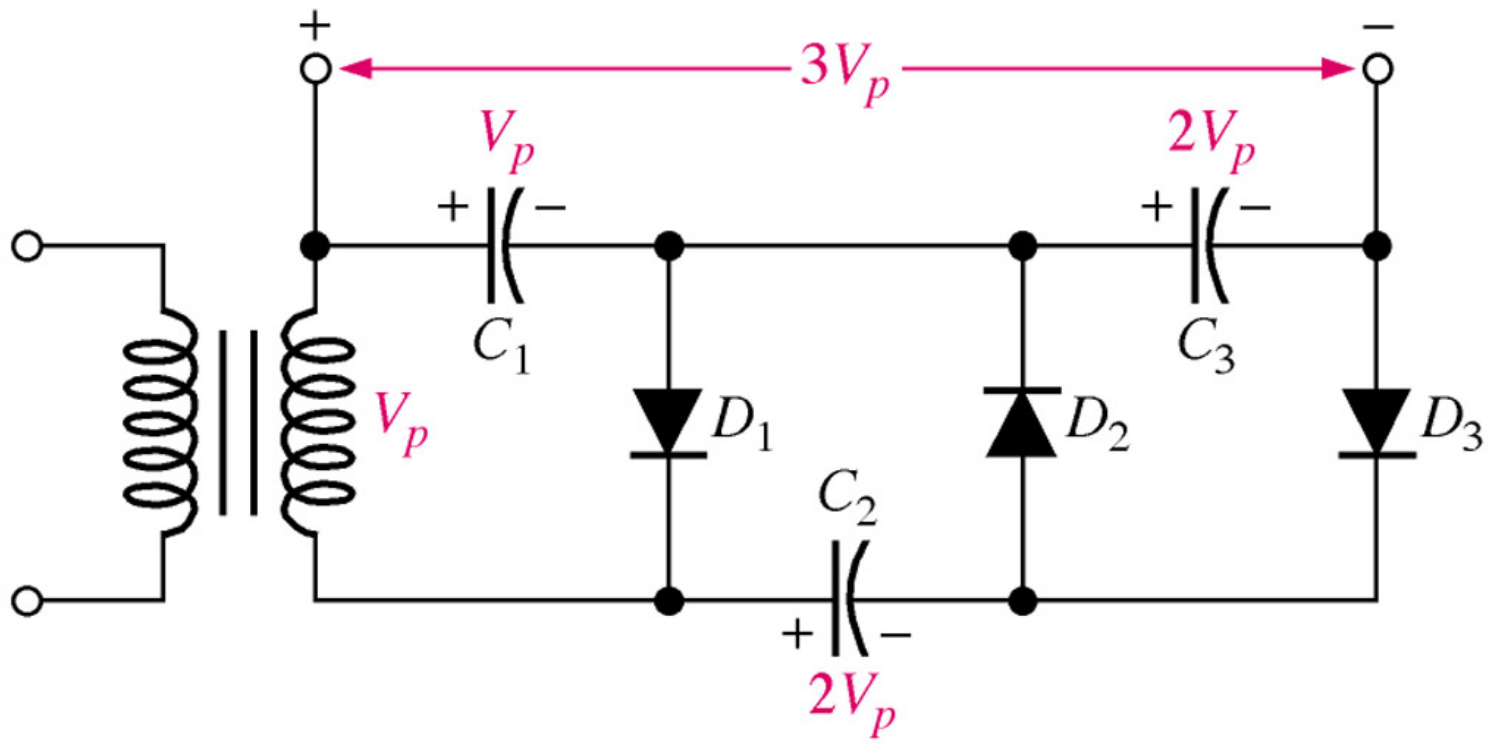
voltage doubler



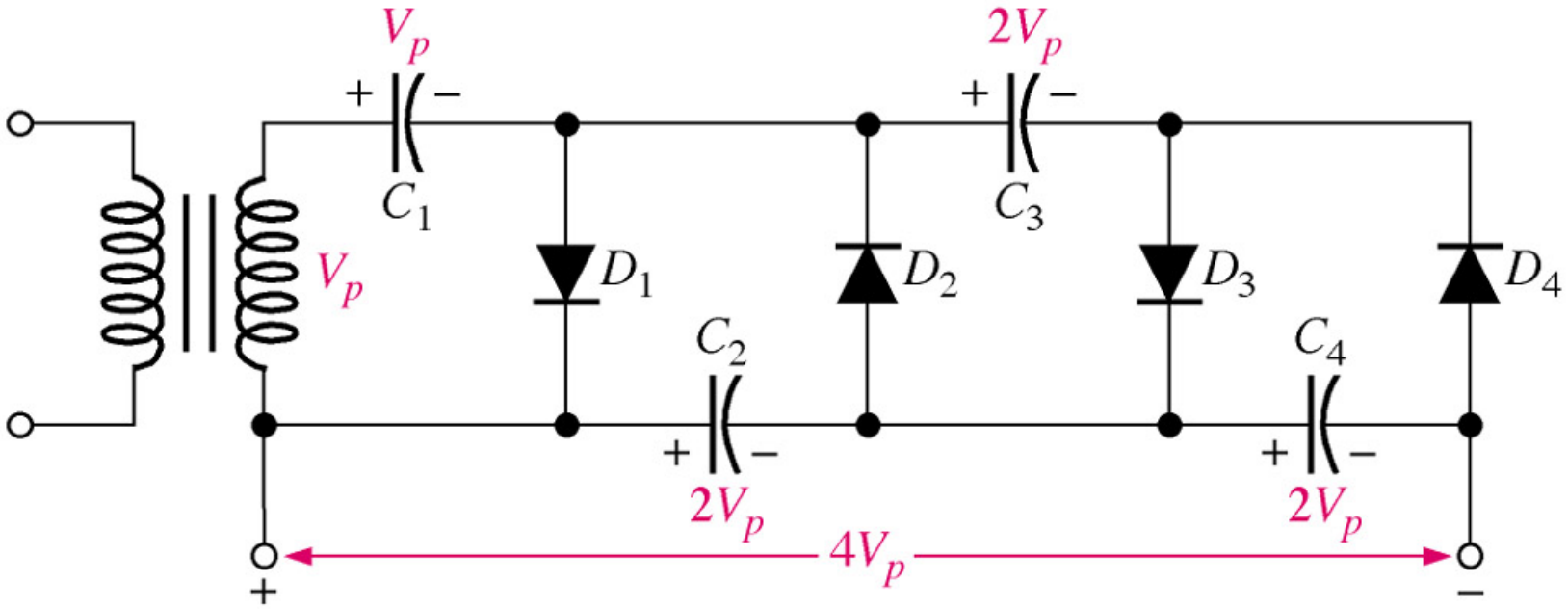
voltage doubler



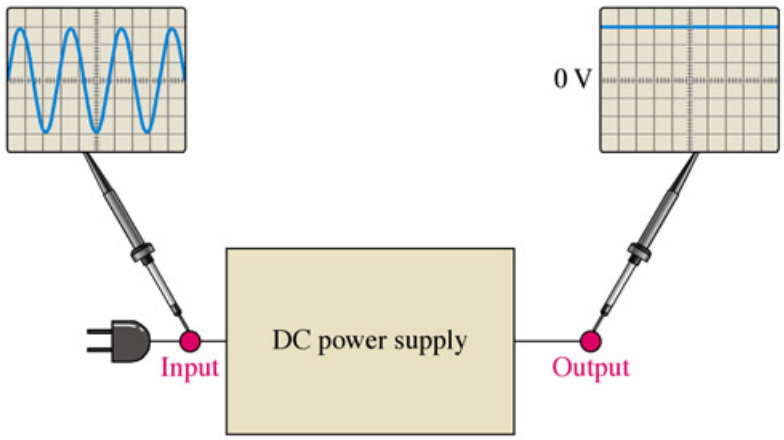
Triple voltage



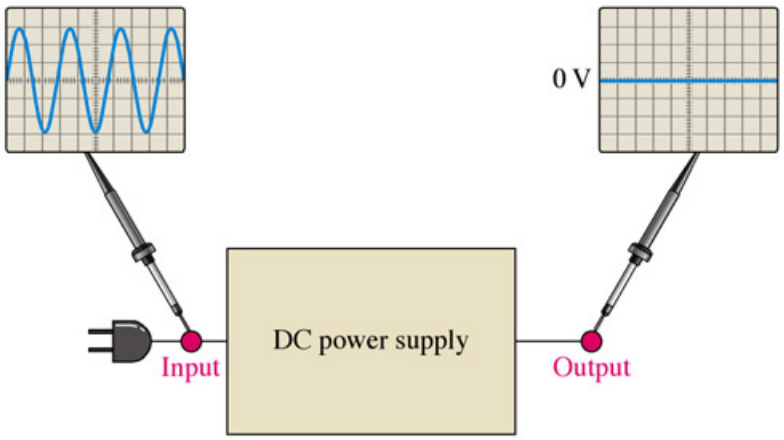
Mạch nhân bốn



Supply voltage

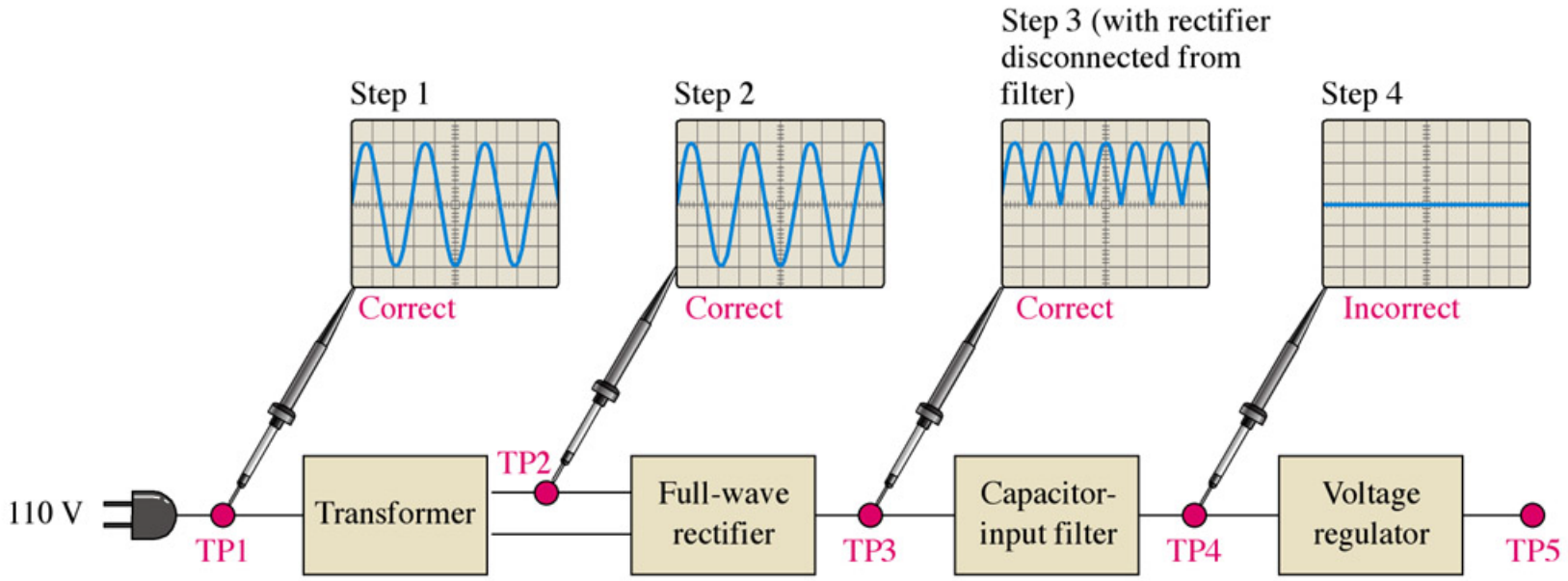


(a) The correct dc output voltage is measured.

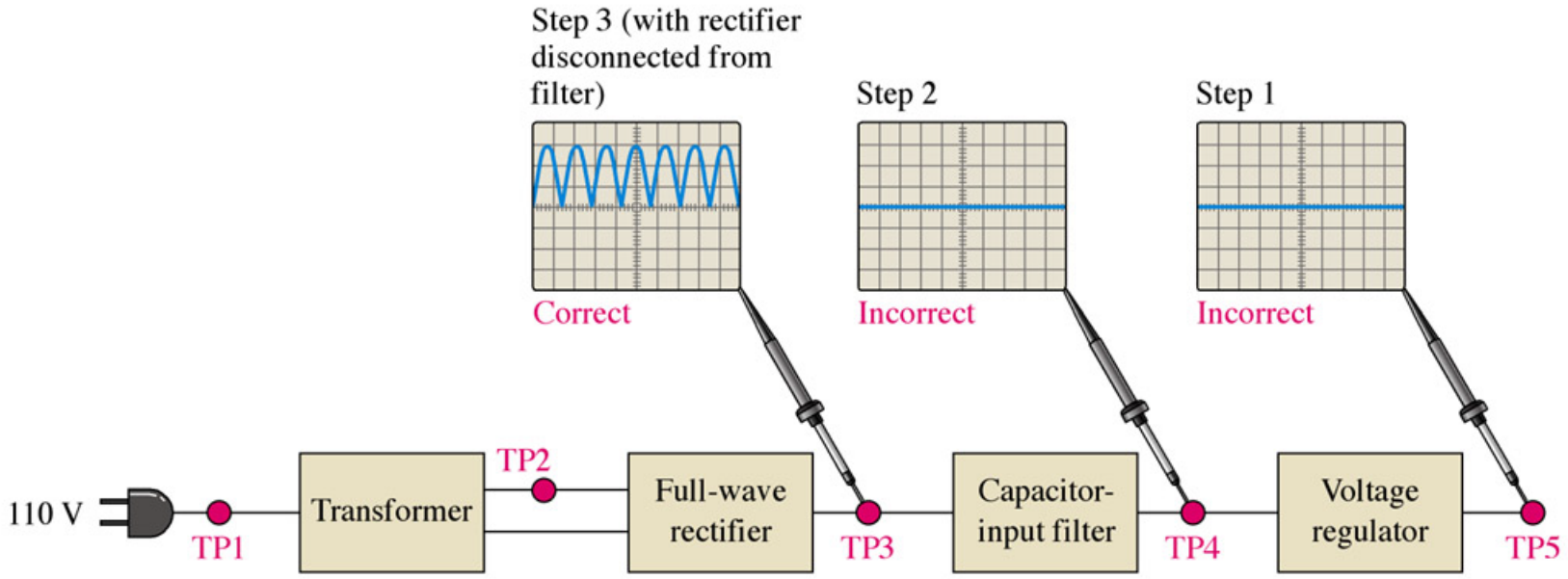


(b) 0 V dc is measured at the output. The input voltage is correct.

Signals at different nodes



Signals at different nodes



Homeworks

- Chapter 2: 1, 5, 6, 10, 11, 15, 21, 23, 24, 27, 30, 34, 37, 42, 47, 49, 52